



Center for Innovation and Security Solutions

Gearbox Artificial Intelligence Tool (GAIT)

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Chapter 1: Preface

1.1 Project Description

Designing aerospace gearboxes traditionally requires long hand calculations, trial-and-error assembly sketches, and several rounds of costly physical tests. The Gearbox Artificial Intelligence Tool (GAIT) project was created to show and prove how an automated workflow can replace most of that effort while still meeting the stringent weight and safety of rotorcraft transmissions.

A compact Python routine takes 31 engineering inputs—power, shaft speeds, material properties, and gear-geometry choices—and computes the initial dimensions of a two-stage helicopter gearbox (spiral-bevel input and planetary output).

The resulting designs are then evaluated in the Romax Concept (Student Edition) software, which validates the weight and size of the Python output result. It delivers accurate mass, safety-factor, and kinematic data for each configuration. From these simulation results we build a fast “prediction engine” that, after a brief training phase, can estimate the total gearbox weight and size for any new set of design parameters in seconds. The prediction engine matches the full Romax Concept (Student Edition) results to within 2% of weight (e.g., 364 kg predicted vs. 357.8 kg simulated) while cutting the evaluation time from hours to a few seconds.

1.2 Project Value

The value of the Gearbox Artificial Intelligence Tool (GAIT) is to harnesses AI to automate the manual design generation and testing processes, creating an intelligent and repeatable workflow that accelerates design development and optimization system and frees engineers to focus on high-value engineering work.

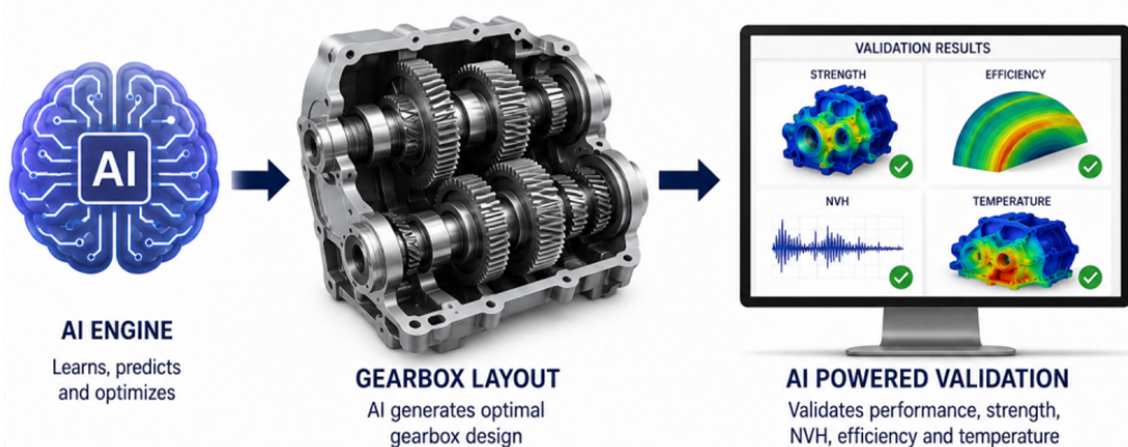


Figure 1.1: GAIT

1.3 Internship Scope

The GAIT project's scope is the following:

- Perform research into the engineering fundamentals and real-world applications of gearbox layout focusing on design and functionality.
- If feasible, evaluate and develop a P-O-C on gearbox parameter optimization using an AI-powered system on an existing gearbox layout.
- Explore the batch processing capabilities of Romax software and evaluate the feasibility of automating gearbox layout validation workflows.

1.3.1 Internship Deliverables

The first phase of the project is the Summer 2026 cohort, with the following deliverables:

- Conduct research and create a detailed report based on the key takeaways.
- Develop an AI system as a Proof of concept: *With initial design parameters from provided resources to estimating optimal weight and size of the gearbox layout.*
- Explore ROMAX and its feasibility for design validation workflow.

1.3.2 Current Project State

The project has successfully established an analytical code parameterized by 31 key design variables, yielding to give the corresponding weight and size of the gearbox design. In this project, 30,000-point dataset was generated based on that code output and using quasi-random sampling to uniformly cover the N-dimensional design space.

Using the generated dataset, HistGradientBoosting Cascade Model was trained successfully to predicts gearbox weight and volume with R^2 score of 0.987, bypassing standard gear snapping discontinuities.

Chapter 2: Introduction

2.1 Introduction to Industrial Gearboxes

Industrial gear reducers are specialized mechanical assemblies designed to alter the rotational speeds and torque profiles between a power source and driven industrial machinery. They serve two fundamental purposes: reducing the shaft velocity and increasing the output torque proportionally. In practice, these sub-assemblies function as critical interface nodes that connect different operating ends:

- **Prime Movers:** High-speed electric motors, internal combustion engines, or steam and fluid turbines.
- **Driven Machinery:** Heavy industrial plant equipment including material conveyors, pulverizing mills, paper processing lines, commercial elevators, liquid mixers, and high-displacement screw pumps.

Standard industrial gearboxes are engineered for mainstream commercial infrastructure that demands reliable operational lifespans and conservative safety margins. To distinguish them from high-performance drivetrains (such as aerospace propulsion or high-speed marine setups), their pitch line velocity is usually restricted to under 25 m/s.

2.2 Practical Challenges in Gearbox Engineering

To an engineer relying solely on textbook mechanics formulas, drafting a gearbox might seem trivial. However, developing a reliable, manufacturable, and cost-effective gearbox requires managing highly coupled, non-linear failure modes. Without practical experience, a designer can easily overlook systemic interactions, causing premature structural failures.

Developing a modern transmission requires balancing multiple coupled and often conflicting performance criteria simultaneously for example:

- **Weight vs. Efficiency and Packaging:** Thinning housing walls minimizes raw material costs and overall mass. However, the resulting reduction in structural stiffness amplifies shaft misalignment. This misalignment triggers uneven gear wear, edge loading, and higher power losses.

2.3 Traditional Gearbox Design

Drivetrain engineering relied on an empirical, handbook-driven methodology. This legacy pipeline leverages standardized physical manuals, historical layout templates, and discrete look-

up matrices to establish a mathematically viable gearbox layout. The workflow operates as a highly sensitive, sequential process executed through manual calculation stages.

2.3.1 System Input Metrics

Before executing any analytical calculations, the master input data matrix must be established from the application requirements.

A synthesis of cross-disciplinary aerospace programs, industrial engineering textbooks, and international gear standards (including NASA's A.R.T. methodologies, AGMA 6013-B14, and ISO 6336) demonstrates a clear consensus on the foundational requirements for manual drive-train synthesis.

Regardless of whether the application demands an ultra-lightweight helicopter transmission or a heavy-duty industrial reduction drive, the absolute core input variables that all engineering papers and design manuals agree upon are:

2.3.1.1 Input Kinetic and Operational Boundaries

- **P:** Nominal input power transmitted by the prime mover (expressed in kW or hp)
- n_{in} : Rotational speed of the high-speed input shaft in RPM
- n_{out} : Target rotational speed of the low-speed output shaft in RPM
- k_a : Empirical application service factor (accounts for prime mover and driven machine shock profiles)
- k_t : Lifecycle duty factor (normalizes operating hours per day against target lifespan)

2.3.1.2 Material Mechanics Inputs

- ρ : Mass density of the component material
- **E:** Modulus of Elasticity / Young's Modulus
- ν : Poisson's ratio
- σ_b : Maximum allowable AGMA root bending fatigue stress
- σ_c : Maximum allowable AGMA surface compressive contact stress
- S_y : Tensile yield strength limit of the shaft material
- S_n : Fully corrected fatigue endurance limit of the shaft material
- **Lubrication and Thermal Limits:** Specific oil viscosity indexes, fluid properties, and targeted operating temperatures required to prevent overheating and maintain oil-film thickness

2.3.1.3 Basic Geometric Setup Choices

- α_n **Standardized Normal Pressure Angle:** The cutting profile orientation, standardized internationally at (20°C or 25°C) that governs contact force direction and controls tooth undercutting limits.
- β **Helix Angle:** The angle of the gear teeth (0°C) for spur gears; typically 15°C to 30°C for helical setups, which introduces axial thrust components into the load paths.
- ψ **Face Width Factor:** The empirical ratio linking tooth width to module size, dictating the width available to spread out mechanical loads. It is a designer-selected constraint variable found by applying strict empirical lookup tables, mechanical stiffness boundaries, and alignment of safety rules.

2.3.2 Kinematic Gear Ratio Determination

The absolute target reduction ratio $u_{overall}$ is calculated from the input and output rotational velocities:

$$u_{overall} = \frac{n_{in}}{n_{out}} \quad (2.1)$$

This value defines the total kinematic reduction ratio. It establishes the exact multiplier by which the rotational input speed drops, and the output torque increases.

2.3.3 Design Torque Calculation

The opening stage of gearbox layout design demands calculating the nominal operating torque at the output shaft ($T_{nominal}$) which is derived from the absorbed torque of the driven machine, or from the nominal output torque of the prime mover (T_{motor}) scaled by the total reduction ratio ($u_{overall}$) and mechanical efficiency (η):

$$T_{nominal} = \frac{P}{n_{in} \left(\frac{2\pi}{60} \right)} \eta u_{overall} \quad (2.2)$$

Where:

- η represents the dimensionless mechanical efficiency coefficient of the gearbox. This factor accounts for internal energy losses turned to heat, rolling resistance inside the shaft bearings, and hydrodynamic churning resistance from the oil lubricant.

To safeguard against unpredictable real-world operating environments, this nominal torque value must be multiplied by an empirical application (service) factor (k_a) and a life duty factor (k_t):

$$T_{design} = T_{nominal} k_a k_t \quad (2.3)$$

2.3.4 Preliminary Center Distance Estimation

Once the design torque is established, engineers estimate the initial center distance between gear axes using empirical output torque curves:

$$\alpha_{estimate} = f(T_{design}) \quad (2.4)$$

This initial baseline geometry provides a starting point before running detailed strength calculations under standards like AGMA or ISO.

2.3.5 Reduction Stage Determination

To preserve reliable gear tooth geometry and avoid overly slender pinions that twist or bend under load, the total reduction ratio must be divided across an optimal number of intermediate reduction stages Table 2.1.

Table 2.1: Ratio Range Selection

Number of Reductions	Economical Overall Ratio Range
Single Reduction	Up to 6:1
Double Reduction	5:1 to 25:1
Triple Reduction	25:1 to 125:1
Quadruple Reduction	125:1 to 625:1

To distribute material costs and torque capacity efficiently, multi-stage gearboxes typically implement a larger reduction ratio at the high-speed input stage, progressively decreasing the ratio toward the lower-speed output stages.

When splitting a total required reduction ratio ($u_{overall}$) across a multi-stage compound gear train (such as a triple-stage reducer), handbook guidelines recommend establishing identical ratio sizing targets across each stage. This balanced approach helps ensure uniform shafting diameters, uniform bearing sizing, and consistent mounting methods throughout the assembly.

To determine the initial target ratio (u_{stage}) for each intermediate reduction step prior to tooth selection, calculate the root of the overall ratio corresponding to the total number of reductions (N_{stages}):

$$u_{stage} = N_{stages} \sqrt{u_{overall}} \quad (2.5)$$

2.3.6 Pinion Tooth Selection and Undercutting Boundaries

The tooth count of the input pinion (z_1) cannot be chosen arbitrarily. To avoid undercutting during manufacturing—which removes material from the tooth root base and severely weakens it—the engineer utilizes the analytical geometry formula matching the selected helix angle (β) and normal pressure angle (α_n). First, the transverse pressure angle (α_t) is determined:

$$\alpha_t = \arctan\left(\frac{\tan(\alpha_n)}{\cos(\beta)}\right) \quad (2.6)$$

Using the transverse pressure angle, the engineer solves for the absolute minimum number of teeth (z_{min}) required to prevent undercutting:

$$z_{min} = \frac{2 \cos(\beta)}{\sin^2(\alpha_t)} \quad (2.7)$$

The engineer consults minimum tooth count reference charts and sets the pinion tooth selection to meet the safety margin.

2.3.7 Parametric Module Trial and Standardization

Next, the framework defines the physical dimensions and structural pitch of the individual gear teeth, quantified as the normal module (m_n). The engineer references minimum tooth count charts to estimate a pinion tooth number (z_1) that avoids undercutting at specified speeds. They evaluate standard analytical formulas to yield a theoretical module:

$$m_{\text{theo}} = \left(\frac{2 T_{\text{design}}}{z_1 \psi \sigma_b Y} \right)^{1/3} \quad (2.8)$$

Where (ψ) is the face width factor selected by the engineer from standard handbook look-up rows and Y is the dimensionless Lewis form factor located manually via handbook tables matching the chosen tooth count (z_1) and pressure angle (α_n).

Because manufacturing tooling is strictly constrained to specific integer intervals, the engineer must consult a Preferred Metric Modules index. The calculated theoretical module is rounded upward to the nearest approved commercial designation.

$$m_n = \text{Standardized}(m_{\text{theo}}) \quad (2.9)$$

Shifting to a standardized module alters the pitch diameters. The engineer must manually re-evaluate the entire geometric stack to compute a modified center distance, completely overriding the initial chart-derived guess:

$$a = \frac{m_n z_1 + m_n (z_1 u_{\text{overall}})}{2} \quad (2.10)$$

Once the standardized module is locked, the true structural face width of the gear teeth is calculated:

$$b = \psi m_n \quad (2.11)$$

2.3.8 Spatial Boundary Audit and Interference Verification

The final phase of the pipeline overlays the calculated component volumes (gear tip diameters, shaft dimensions, and raw bearing widths) directly onto the vehicle's structural envelope. The engineer evaluates the absolute outermost tip diameter of the largest gear assemblies:

$$D_{\text{tip}} = m_n (z_2 + 2) \quad (2.12)$$

They physically verify if the rotating elements clear the static airframe frame and fit within the designated housing box.

2.3.9 AGMA Gear Tooth Bending Failure Analysis

Once the standardized geometry is finalized, the engineer evaluates the mechanical stresses at the tooth contact points to ensure safety margins are met. To calculate the stress concentrations pulling at the base of the tooth root, the tangential load (F_t) is determined:

$$F_t = 2 \cdot \frac{T_{\text{design}}}{d_1} \quad (2.13)$$

The complete AGMA root bending stress (σ_b) equation incorporates dynamic adjustments for operational variables:

$$\sigma_b = \frac{F_t K_a K_v K_m}{b m_n J} \quad (2.14)$$

Where:

- K_a : External application shock factor
- K_v : Dynamic velocity multiplier tracking tooth profiling errors
- K_m : Load distribution factor mapping shaft misalignment
- b : Structural face width of the gear mesh
- J : AGMA geometry factor for root bending

The safety criteria requires that the calculated stress remains below the material's allowable threshold.

2.3.10 AGMA Surface Compressive Contact Fatigue Analysis

To protect the tooth profiles from compressive contact pitting, the Hertzian contact stress (σ_c) across the fluid film interface is calculated:

$$\sigma_c = Z_E \sqrt{\frac{F_t K_a K_v K_m}{b d_1 I}} \quad (2.15)$$

Where:

- Z_E : Elasticity coefficient matching the combined material moduli
- I : AGMA geometry factor for surface contact pitting resistance

2.3.11 Dual-Shaft Fatigue Sizing via the ASME Elliptic Criterion

With the center distance locked, the engineer sizes the solid support shaft segments (D_1, D_2) to resist catastrophic failure under combined loading conditions. The gear tooth loading vectors generate a severe alternating bending moment (M) along the shaft, while the rotation transmits a steady torsional shear torque (T). To calculate the absolute minimum safe shaft diameter ($d_{\min}$) required under a design safety factor (N), apply the ASME Elliptic Fatigue framework:

$$d_{\min} = \left(\frac{32 N}{\pi} \sqrt{\left(\frac{K_f M}{S_n} \right)^2 + \frac{3}{4} \left(\frac{T}{S_y} \right)^2} \right)^{1/3} \quad (2.16)$$

Where:

- K_f : Fatigue stress concentration factor

The manual shaft dimensioning must satisfy this sizing boundary:

$$D_{\text{shaft}} \geq d_{\min} \quad (2.17)$$

2.3.12 Sizing Limitations and Analytical Realities

These macro-sizing calculations provide only a preliminary estimation of the gearbox envelope under idealized conditions. In practice, this represents a highly simplified starting point; an engineer cannot rely solely on these broad formulas to finalize a design. Transitioning to a field-ready unit requires extensive multi-physics iterations, as localized tooth stresses, dynamic misalignments, and housing flex will fundamentally alter the system's real-world performance. A summary of the stages of the gear design workflow is shown in Figure 2.1.



Figure 2.1: Gear Design Workflow

While this baseline workflow suffices standard commercial gearboxes, it falls short in high-performance sectors like aerospace, where weight and acoustic constraints are severe. To push past these classical boundaries, specialized research initiatives are required to redefine standard optimization rules. A prime example of this transition from standard commercial sizing to advanced drivetrain engineering is the Advanced Rotorcraft Transmission (A.R.T.) program.

Chapter 3: Advanced Rotorcraft Transmission (A.R.T.) program

3.1 Introduction

The Advanced Rotorcraft Transmission (A.R.T.) program was created to develop and demonstrate lightweight, quiet, durable drive train systems for the next generation of rotorcraft. Sikorsky is one of the participants, and this program allows them to evaluate key emerging material and component technologies and novel design concepts for advancing the technology for future rotorcraft transmission.

One of the specific objectives that is relevant to the project is “Reduce transmission weight by 25 percent relative to SOA — the most recent stage in the development of a product, incorporating the newest technology, ideas, and features — trends (currently in the range of 0.40 lb./hp)”. A.R.T. is designed to provide a high return on weight improvement by developing novel concepts involving high risk factors.

The A.R.T program requires all participants to select a vehicle fitting into one of the two distinct next-generation aircraft classes and to demonstrate their proposed advanced-concept drive trains. The first one is Future air attack vehicle (FAAV): a 10,000- to 20,000-pound aircraft capable of undertaking tactical support and air-to-air missions, and second is advanced cardo aircraft (ACA): a 60,000- to 80,000-pound aircraft capable of heavy-lift field support operations.

To evaluate the effect of advanced-technology improvements on aircraft size and performance, computer programs were initially used to develop the reference aircraft design. In the early stage of this program, a few key parameters were used to develop the drive system design. As the design progressed, detailed rotor and engine performance data were entered to improve the precision of the calculations. The program was used to discover broad trends early in the design process, and it was also used to determine the effect of relatively small modifications once the design was more developed.

Included in the chapter is also the trade studies selection process. As during the early development of the A.R.T. drive system, comparisons and trade studies were conducted among novel transmission configurations with high payoff potential.

3.2 Program Objectives

While standard industrial gearboxes prioritize manufacturing cost and traditional safety factors, aerospace drivetrains demand ultra-lightweight, quiet, and highly durable systems. The Advanced Rotorcraft Transmission (A.R.T.) program—a joint initiative funded by the U.S. Army and NASA—was established to drive drivetrain technology beyond historical baseline limits. The program set strict performance targets to improve on contemporary baseline aircraft metrics:

- **Weight Reduction:** Lower total transmission weight by **25%** relative to prevailing baseline configurations, which typically trended in the range of 0.40 lb/hp.
- **Acoustic Attenuation:** Suppress operational noise levels by **10 dB** at the transmission source, reducing acoustic fatigue on components and passengers.
- **Operational Reliability:** Extend the Mean Time Between Removals (MTBR) to a minimum of **5,000 hours** to lower lifecycle costs and enhance mission readiness.

Drivetrain weight reductions have direct, compound benefits for overall aircraft performance. Retrofitting a baseline helicopter with an advanced A.R.T. transmission allows saved empty weight to be converted into useful load capacity without increasing gross takeoff weight. This capacity can be allocated directly to cargo payload or fuel volume. Increasing fuel capacity extends the aircraft's operational radius of action, allowing for longer flight profiles and enhanced mission flexibility.

3.3 Advanced Structural Drivetrain Configurations

To meet these targets, contractors evaluated and validated several advanced architectural configurations:

1. Self-Aligning Bearingless Planetary (SABP) Configuration

This layout replaces conventional, heavy planet roller bearings and heavy carrier assemblies with an internal design utilizing flexible structural rings. By allowing the planet gears to float and self-align within the mesh, the system automatically balances tooth loads across all paths. This architecture eliminates high-wear bearings, lowers structural weight, and improves system reliability.

2. Split-Torque Configuration

The split-torque design divides the incoming power from a single engine input into multiple low-power, parallel gear paths. These separate branches recombine a single output gear. Splitting the torque allows for smaller gear faces and narrower tooth profiles, which reduces total system weight and provides redundant load paths.

3. Conventional Single-Stage Planetary Configuration (Aerospace Optimized)

This approach refines the standard planetary layout using advanced high-strength steel alloys and optimized micro-geometry. It maintains central sun gear, external stationary ring gear, and a planet carrier. This baseline configuration was used as a benchmark to measure the weight and noise advantages of alternative architectures.

3.4 Program Development-Sikorsky

The major objective of the A.R.T. contract was the development of a high reduction ratio, torque splitting output stage. For minimum weight, the output stage of the split path design has a high aspect ratio of face width to diameter.

“Preliminary designs were conducted of split path and split torque transmissions to evaluate the weight, reliability, and noise and compare the improvements to the program goals. A split torque gearbox with a high reduction ratio double helical output gear was determined to be 23the baseline two stage planetary design. The improvements were attributed to extensive use

of composites, spring clutches located at the transmission input, advanced high hot hardness gear steels, the split path configuration itself, high reduction ratio, double helical gearing on the output stage, elastomeric load sharing devices, and elimination of accessory drives.”

Baseline gearbox weight was determined using statistical weight equations that utilize preliminary weight parameters such as design torque and reduction ratio. Layout weight calculations were conducted on the advanced split path and split torque designs to obtain the individual component weights and to enable identification of bulk items which the layout does not include such as hardware, protective coatings, sealants, bosses on castings, lubrication system components, oils, etc. Once identified, the bulk item weights are estimated using values obtained from systems of similar design. These layout calculations are compared to the analytical weight and differences resolved.

Since one engine inoperative (OEI) power is felt all the way to the final stage bull gear mesh in the split path type of arrangement, the gearbox is essentially sized to OEI conditions.

A cross-sectional layout of the ACA baseline main gearbox having the configuration of the CH53E is depicted in Figure 3.1 which also shows the rpms, reduction ratio's, numbers of teeth, and pitch diameters for the main meshes. This gearbox has been sized to the speed and power requirements of the ACA.

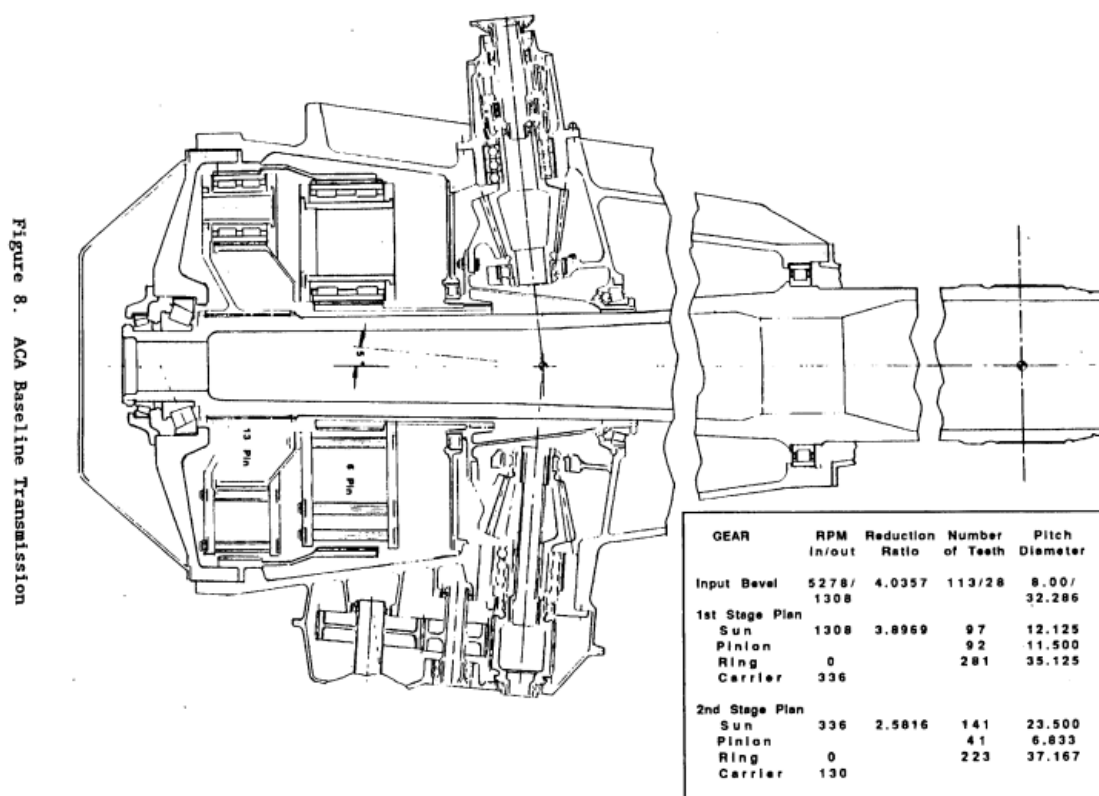


Figure 3.1: ACA Baseline Transmission

3.5 Approach and Methodology

The A.R.T. weight savings results in additional reductions when the ACA aircraft is downsized for the same payload and performance. Weight savings from the transmission alone is 2,508 pounds. If the aircraft size is not changed, the payload will increase by the transmission weight savings. If the payload is kept constant, the aircraft can be downsized.

Note that the weight empty is reduced by 2,508 pounds for the baseline w/A.R.T. since this is the savings for the A.R.T. transmission system alone. There is however an additional 1,201 pounds savings by downsizing the aircraft for the same payload thus the ART has actually saved 3,709 pounds in the aircraft. When resizing the aircraft for the same payload, certain aircraft design parameters were kept constant. The engines, three GE-38's, were fixed for all designs. The cabin size was fixed. The payload/range was kept at 25,000 pounds with a radius of action of 500 kilometers. The takeoff criterion of 200 feet per minute vertical rate of climb at 95 constant. With the disk loading maintained at approximately 10 pounds per square foot, the main rotor diameter was sized to match the power required with the power available at the takeoff condition.

Table 14. Weight Savings by Downsizing Aircraft for Same Payload

Parameter	Baseline	Baseline w/ART	Downsized w/ART
Gross Weight	85,764	85,764	81,957
Payload	25,000	27,508	25,000
Mission Fuel	15,316	15,316	15,218
Weight Empty	45,448	42,940	41,739

Figure 3.2: Weight Saving by Downsizing Aircraft for Same Payload

All aircraft design analysis was accomplished with the helicopter design model (HDM). The HDM is a computer program which uses basic input data for mission requirements, design criteria, configuration, engine, and technology. The program then calculates dimensions, weights, powers, performance parameters, and costs. The HDM uses experience factors for calculating many of the outputs. For example, weight trending equations are provided for the main gearbox, main rotor shaft, main rotor blades, intermediate gearbox, tail gearbox, and tail drive shafting. These equations use parameters such as horsepower, rpm, and reduction ratio with coefficients derived from previous designs to calculate weights.



Figure 3.3: Helicopter Design Model

3.6 Trade Studies

Four preliminary designs were completed in the preliminary design phase of the program. Designs of split path and split torque gearboxes were completed for a high output bull gear reduction ratio of 10.85:1 (141/13) and a lower ratio of 8.15:1 (106/13) for a total of two ratios of split torque and two ratios of split path designs.

When they entered the detail design phase, more realistic calculations revealed that the high ratio configuration (10.85:1) produced excessive stresses in the bull pinion shaft (the shaft that drives the big “bull” gear). To stay within safe stress limits, the maximum ratio was retuned down to 9.77:1 (which is the ratio 127/13).

Split-Torque uses a planetary/“differential” arrangement to split the engine’s torque before it goes to the main gearbox. A differential type planetary gear set splits the torque that comes from the two engines. Split-Path has two engine drives which go through separate paths that meet only at the final “bull” gear that delivers power to the rotor. This means that it runs the two engine drives on separate mechanical paths that only meet in the very last gear (the bull gear).

The trade-off analysis ultimately selected the split-path gearbox powered by a lower reduction ratio (106/13) because it is the only architecture that simultaneously delivers the lightest weight, the highest reliability (MTBR), room-for-growth in manufacturing, and compliance with the low-noise goal. The split-torque design, despite its inherent torque balance, fell short due to extra gear stages, heavier bearing loads, and reduced redundancy; all of which would have increased weight and lowered MTBR for the ACA’s three-engine configuration.

Chapter 4: Gearbox

4.1 Introduction

Gear-driven components: turboprop engine propellers, helicopter main and tail rotors, and various engine auxiliaries coordinate the speeds of power-consuming shafts with the engine rotor. Because propellers and most auxiliaries run at lower speeds, a gearbox (reduction drive) is required to reduce the engine's peripheral rotation. While basic gearbox design principles are taught in "Mechanics," aircraft gearboxes possess distinctive characteristics: they demand higher cooling intensity due to increased oiling; operate in multi-mode, shortening their service life; face challenges maintaining precise steady modes, especially for turboprop (TPE) and turboshaft (TShE) engines due to torque-metering unit difficulties; suffer installation complications at the engine front, affecting velocity triangles and stressing the compressor face; and must be as light as possible.

4.2 Gearbox Limitations

The most crucial parameters of the gearbox are mass and diameter. Mass of modern TPE gearboxes makes up 25–30 % of the overall engine mass. The gearbox of a helicopter is usually two or three times heavier than the TShE itself. The greater the transmitted power and the gear ratio, the heavier the gearbox is. The specific mass of the TPE gearbox (ratio of the gearbox mass to the power transmitted) ranges within $0.05\text{--}0.1 \text{ kg kW}^{-1}$.

Diametrical size must be minimal. This is of great importance for the gearbox inside the engine. If the gearbox diameter exceeds the diameter of the compressor, it is very hard to guide air to axial compressor inlet with minimal hydraulic losses.

Thus, the task for the engineers is to design a light, reliable, and small gearbox able to transfer power at high efficiency. The problem becomes complicated if the gearbox is expected to transmit great power values or when its gear ratio is high.

The gearboxes driving a single lift rotor or double lift rotor are designed using simple, planetary and differential gears. They can be single-stage or multiple-stage depending on gear ratio.

4.3 Classification of Gearbox

Gearboxes can be classified by different features:

1. Number of input drive shafts

Conventional gearboxes have one input and one output shaft. However, designers often use alternative configurations: a single input with two (usually coaxial) outputs for high-power engines (e.g., NK-12); a branched diagram with two outputs for helicopters with

a lift and tail rotor; a gearbox with two inputs and one output when two turboprop engines drive a single propeller; and a single gearbox serving two engines (e.g., Mi-24's VR-24) that powers lift and tail rotors or coaxial lift rotors (Ka-25). Thus, gearboxes can feature combinations such as two inputs and two outputs, as seen in the RV-3F driven by GTD-3F engines.

2. Shaft arrangement

Turboprop engines use coaxial gearboxes. In twin-engine configurations, the gearbox uses two parallel input shafts, whereas gearboxes with two intersecting shafts are used exclusively in helicopter power plants.

3. Number of stages

Gearboxes may comprise one, two, three, or four stages, with the four-stage configuration reserved for helicopter power plants. The arrangement of the stages can vary between designs.

4. Kinetics of the stage

Stages of gearboxes can be grouped into three types: simple, planetary, and differential. Differential stages are further divided into loop (single degree of freedom) and two-degree-of-freedom variants.

5. Gear type

Most gearboxes use spur gears of diverse sizes, while bevel wheel trains appear only in configurations with crossing input shafts, such as the gear trains found in helicopters.

4.4 Fundamentals for Designing the Gearbox

4.4.1 Choose the Gear Type

- **Spur Gears**

Simple gears with straight teeth. Best for low-speed, high-torque applications.

- **Helical Gears**

Teeth are cut at an angle, allowing for smoother and quieter operation. Used for higher speed applications.

- **Bevel Gears**

Used for transmitting power at a 90 degree angle.

- **Worm Gears**

A gear arrangement where a worm (a gear with a screw-like thread) meshes with a worm wheel, providing high reduction ratios and compact design.

- **Rack and Pinion**

Used to convert rotational motion into linear motion.

4.4.2 Determine Gear Ratio

The gear ratio is crucial for controlling the speed and torque at the output shaft. The formula for calculating the ratio will differ depending on the type of gearbox arrangement. The formula for a worm gearbox is different for that of an epicyclic gearbox.

$$m = \frac{T_G}{T_P} \quad (4.1)$$

Equation 4.1 represents a single stage gearbox and would be ideal when designing a worm gearbox or a compound gear train. A note to keep in mind is a higher gear ratio provides more torque but reduces speed, while a lower ratio gives higher speed with less torque.

$$m = \frac{G_2}{G_1} \cdot \frac{G_4}{G_3} \cdot \frac{G_6}{G_5} \quad (4.2)$$

Equation 4.2 would be applied for a multistage compound gearbox. The three stages represent the reduction between each set of shafts. Another important aspect to keep in mind is maintaining the size and weight of the gearbox. For example, a total reduction ratio of 15:1 is desired to reduce the speed of 3000 rpm.

$$n_{\text{output}} = \frac{n_{\text{input}}}{m} = \frac{3000 \text{ rpm}}{15} = 200 \text{ rpm} \quad (4.3)$$

With this information, two factors are known, the input speed, and what is required for the output speed. The next phase of developing the gear train is keeping each gear as small as possible while still meeting the speed and strength requirements. This can be achieved for a gearbox by taking the cube root of the gear ratio as shown in Equation 4.4. The cube root is being used because this example is a three-stage gear reduction.

$$m_{\text{ea}} = \sqrt[3]{m} = \sqrt[3]{15} = 2.47 \approx 2.5 \quad (4.4)$$

The value produced by Equation 4.4 should be the targeted gear ratio between each of the three stages. Given that it is not a whole number, the value may need to be rounded up or down depending upon the output speed requirements. If a specific stage must be adjusted to a larger or smaller ratio, that will be fine, but this serves as a good starting point in the development. Following this procedure will allow for having the same gears in each ratio, which will mean similar shafting, bearings and mounting methods.

4.4.3 Material and Component Selection

The material selection for gearbox components is a critical engineering decision that must account for the distinct functional requirements of each element, ranging from aluminum housings and steel bearings to medium-carbon steel shafts and exposed gears. The applicability of a particular material type depends on load conditions, expected service life, cost constraints, and fabrication feasibility, necessitating meticulous evaluation of mechanical properties, corrosion resistance, and manufacturability for each part. Common gear materials include high-strength steel, which offers excellent durability for high-load applications; advanced alloys, which provide lightweight yet high-performance characteristics suitable for aerospace and other demanding sectors; bronze, whose inherent wear resistance makes it the material of choice in

worm-gear configurations; and engineering plastics, which are often deployed in lower-load or noise-sensitive environments. Each material class must be carefully matched to its respective component to achieve optimal performance and longevity of the overall gearbox system.

4.5 Multi-stage Gear Transmission Design

The objective functions considered in this work are geartrain efficiency and total weight, aiming to design a transmission that maximizes efficiency while minimizing mass.

4.5.1 Geartrain Weight

The overall volume of the system is approximated as the sum of the volumes of each gear. Rather than modeling each gear tooth explicitly, the gear is represented as a disk with an outer diameter equal to the pitch diameter and an inner diameter corresponding to the defined bore. In this simplified approach, the volume of each gear is computed as the product of its cross-sectional area and face width. The total weight of the transmission can be calculated as:

$$W = \rho_{\text{steel}} V = 7.8 \times 10^3 \text{ kg/m}^3 \sum_{i=1}^n \pi b_i \left[\left(\frac{m_i z_i}{2} \right)^2 - r_{\text{bore}}^2 \right] \quad (4.5)$$

This assumption approximates the bulk geometry while decreasing computational effort.

Volumetric Approximation: In preliminary optimization, the total mass is estimated by summing the volumes of each gear, modeled as disks. The formula used is:

$$\text{Total Weight} = \sum (\text{Cross-sectional Area} \times \text{Face Width} \times \text{Material Density})$$

This approach approximates the bulk geometry to decrease computational effort during design iterations.

4.5.2 Design Constraints

The constraints applied in this study are divided into three categories: geometric, layout, and safety constraints. Geometric constraints relate to the dimensions, form of individual gears, and ensure the feasibility of each gear stage. Layout constraints address spatial and positional relationships between multiple gear stages, maintaining compatibility within the overall transmission architecture, and satisfying specific requirements. Safety constraints concern the strength of the gears and are assessed through ISO standardized safety factors.

4.5.3 Safety Constraints

The root and flank safety factors are the ratios between the modified value of allowable bending (σ_{at}) and contact (σ_{ac}), and the calculated bending (σ_t) and contact (σ_c) stress as shown in Equation 4.6 and Equation 4.7. The modification or service factors are stress cycle factor (Y_N) for bending strength, which is imposed 0.7 for idler and 1 for simple gears, temperature factor (Y_σ), reliability factor (Y_Z), stress cycle factor (Z_N) for contact stress, and hardness ratio factor (C_H) for contact stress.

$$S_F = \frac{\sigma_{at} Y_N}{\sigma_t Y_\sigma Y_Z} \geq S_{F,\min} \quad (4.6)$$

$$S_H = \frac{\sigma_{ac} Z_N C_H}{\sigma_c Y_\sigma Y_Z} \geq S_{H,\min} \quad (4.7)$$

The calculation of bending stress (v_t) and contact stress (v_c), as well as the corresponding unmodified allowable stress values, follow the procedures outlined in ISO 6336. These values are derived based on the applied tangential load, gear tooth geometry, and material properties.

5.1 Helicopter Transmission Architectures and the UH-60 Black Hawk MGB

In rotorcraft engineering, the Main Gearbox (MGB) is the mechanical heart of the aircraft. It must safely transmit thousands of kilowatts of power from high-speed gas turbine engines (typically running between 6,000 and 21,000 RPM) down to the slow, extremely high-torque operational speeds dictated by the main rotor blades (typically 250 to 400 RPM). To achieve this immense reduction while remaining light enough for flight, helicopter transmissions universally employ a multi-stage architecture.

The standard, most reliable MGB configuration consists of a horizontal double-input engine stage transitioning through a 90° spiral bevel reduction into a vertical, high-torque final planetary (epicyclic) stage as shown in Figure 5.1.

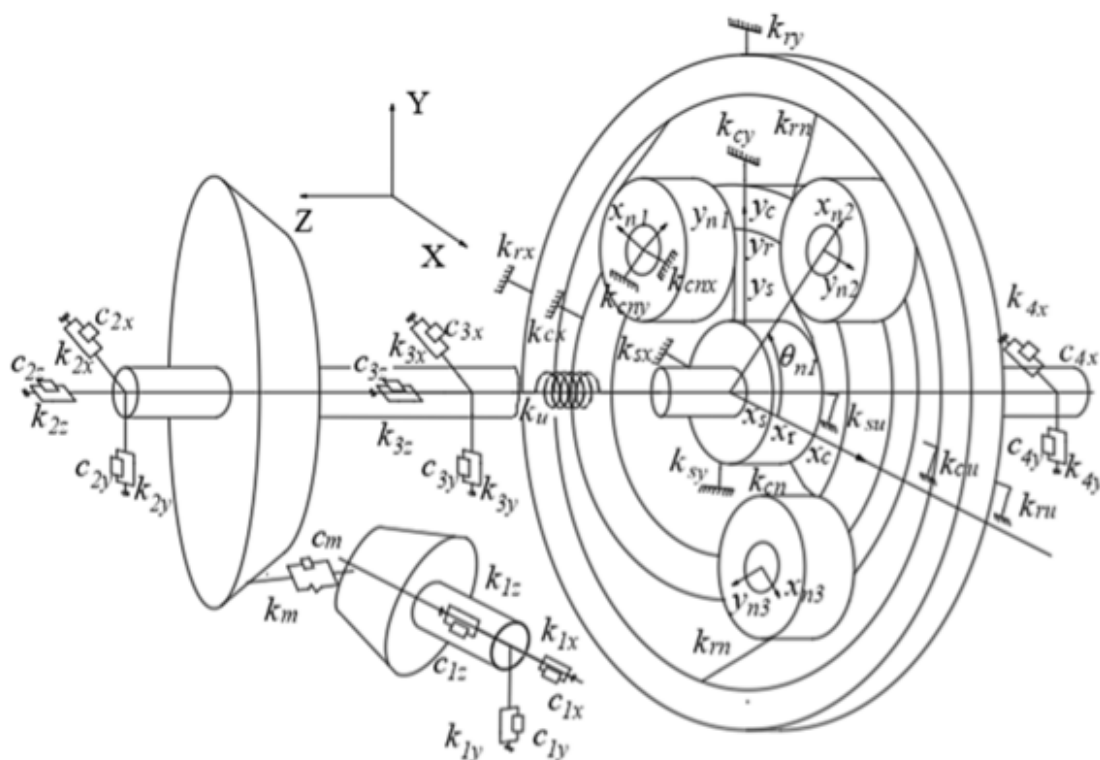


Figure 5.1: Helicopter MGB Configuration

A prime operational example of this configuration is found in the Sikorsky UH-60 Black Hawk. The Black Hawk uses a modular, triple-stage main transmission. Power enters horizontally from two General Electric T700 turboshaft engines into two independent input modules.

Each module uses high-speed helical gears to combine the engine inputs, which then drive a Spiral Bevel Gear mesh. This bevel mesh executes a 90-degree axis flip, turning the horizontal engine power into a vertical spinning axis. This vertical core shaft feeds directly into a final, single-stage planetary gear system located beneath the main rotor mast. The planetary stage uses a fixed outer ring gear and five power-splitting planet gears mounted to a forged steel planet carrier, which directly rotates the main rotor shaft to lift the aircraft.

5.2 Sizing Input Parameters

To build an MGB, every baseline physical input, material limit, and structural factor must be explicitly stated. The following section defines the variables required, along with their standard engineering ranges for aerospace and heavy-duty drivetrain applications:

1. Operational Inputs

- **P [W] (Nominal Input Power):** The total rated mechanical power delivered by the driving gas turbine engine modules.
- n_{in} [RPM] **(Input Engine Speed):** The rotational speed entering the horizontal input module. Typical turboshaft outputs range from 6,000 to 21,000 *RPM*.
- n_{out} [RPM] **(Target Main Rotor Speed):** The final required speed of the rotor mast, strictly bounded by blade radius aerodynamics to prevent supersonic tip shockwaves. Typical limits range from 250 to 400 *RPM*.
- N_p **(Number of Planet Gears):** The integer count of power-splitting planet channels evenly spaced inside the ring gear. Typical configurations utilize 3 to 6 planets; the UH-60 Black Hawk utilizes 5 planets.

2. Material Strength and Density Limits

- $\sigma_{allow,F}$ [Pa] **(Allowable Bending Stress Limit):** The maximum tensile fatigue stress the tooth root can handle over millions of cycles before fracturing. For aerospace-grade carburized steels, this ranges from 400 MPa to 550 MPa.
- $\sigma_{allow,H}$ [Pa] **(Allowable Contact Stress Limit):** The maximum baseline surface pressure the tooth faces can handle before pitting. For case-hardened aerospace steels, this ranges from 1,400 MPa to 1,750 MPa.
- $\sigma_{allow,shaft}$ [Pa] **(Allowable Shaft Fatigue Limit):** The fully modified endurance limit of the shaft metal under reversed loading. Typical bounds range from 100 MPa to 180 MPa.
- Z_E [[Pa^{0.5}]] **(Material Elastic Coefficient):** Combines Young's Modulus and Poisson's ratio to govern micro-contact strain flattening.
- ρ_g [kg/m³] **(Gear Material Density):** High-strength carburized alloy gear steel matches a hard constant of approximately 7,850 kg/m³.
- ρ_s [kg/m³] **(Shaft Material Density):** Forged steel structural shafts match a constant of approximately 7,850 kg/m³.
- ρ_h [kg/m³] **(Housing Material Density):** To minimize flight mass, helicopter casings utilize ultra-lightweight magnesium alloys 1,800 kg/m³ or high-grade aluminum alloys 2,700 kg/m³.

3. Geometric and Structural Factors

- α_n [**Degrees**] (**Normal Pressure Angle**): The base angle cut of the gear profile teeth. Standard industrial hobs default to (20°C or 25°C).
- β [**Degrees**] (**Helix Angle**): The angle of the helical teeth cut relative to the shaft axis. Standard industrial limits range from (10°C to 30°C); a setting of (0°C) reduces the algorithm to a basic spur profile.
- λ_d (**Planetary Face-Width-to-Diameter Ratio**): The chosen aspect ratio of the gear width relative to its sun diameter. To maintain structural stiffness against shaft deflection, standard practice enforces 0.5 – 2.0.
- λ_b (**Bevel Face Width-to-Diameter Ratio**): Aspect ratio bounding the spiral bevel stage cone width. Standard Aerospace Range: 0.25 to 0.35.
- I_b and I_p (**AGMA Geometry Factors**): Dimensionless indices mapping localized involute curvature profiles for the planetary and bevel teeth respectively. Planetary Range: Evaluated via ratio kinematics (0.11 to 0.15). Spiral Bevel Range: Catalog/standard fixed constant (0.07 to 0.085).
- C_w (**Weight Correction Factor**): The percentage multiplier modelling material precision-machined out of the gear blank core to create lightweight flight webs. A solid blank is 1.0, while pocketed aerospace blanks range from 0.4 to 0.7.
- C_t (**Ring Wall Thickness Factor**): The ratio of the ring gear outer wall thickness relative to the standard module size, sized to prevent radial hoop fatigue. Valid boundaries enforce 3.5 to 6.0.
- C_c (**Carrier Plate Width Factor**): Sizes the width of the structural plates flanking the gears as a percentage of the face width. Standard packaging balances between 0.2 and 0.4.
- K_a (**Application/Overload Factor**): A designer-selected safety cushion multiplier that increases sizing limits to handle sudden engine torque surges or aerodynamic rotor blade shocks. Standard boundaries dictate 1.2 to 1.5.
- K_t (**Temperature Factor**): An environmental fatigue scaling modifier. Set to a constant of 1.0 for all operational gearboxes running under a thermal ceiling of (120°C).
- n_s (**Shaft Factor of Safety**): The explicit safety margin index applied to the shaft diameters to handle multi-plane loading uncertainty. Standard practice ranges from 1.5 to 3.0.

4. Geometric and Structural Factors

- t_{case} [m] (**Perimeter Wall Thickness**): The casting wall thickness of the main outer gearbox body barrel shell. Standard bounds measure between 6 mm to 14 mm.
- t_{plate} [m] (**Face Plate Thickness**): The structural thickness of the casing end caps enclosing the assembly. Standard bounds range between 10 mm and 20 mm.
- $L_{clearance}$ [m] (**Assembly Clearance Space**): The baseline physical gap length allowed on each end of the gear stacks to clear supporting oil seals, retaining snap-rings, and bearing inner caps. Standard spacing defaults between 20 mm and 50 mm.

5.3 Kinematic Speed Mapping

To find the weight and size of the MGB, the entire transmission is modelled as a unified structural closed loop. Sizing begins by tracing how torque and rotational velocities scale across the system zones based on nominal input power (P), input engine speed (n_{in}), bevel reduction ratio (u_b), and planetary reduction ratio (u_p).

The total kinematic reduction ratio ($u_{overall}$) is the strict product of both internal stages:

$$u_{overall} = u_b \cdot u_p = \frac{n_{in}}{n_{rotor}} \quad (5.1)$$

The intermediate velocity on the vertical transition shaft between the bevel output and planetary input is calculated automatically once the optimal bevel ratio is resolved by the global optimization loop

$$u_{vertical} = \frac{n_{in}}{n_b} \quad (5.2)$$

The nominal torque profiles at each independent checkpoint are established by applying the AGMA application overload factor (K_a) and temperature factor (K_t):

$$T_{\text{design bevel}} = \frac{30 P}{\pi n_{in}} \cdot K_a \cdot K_t \quad (5.3)$$

$$T_{\text{design planet}} = \frac{30 P}{\pi n_{in}} \cdot K_a \cdot K_t \cdot \eta_b \cdot u_b \quad (5.4)$$

Where η_b represents the localized mechanical transmission efficiency of the spiral bevel mesh.

5.4 Mathematical Derivation of Component Weights

To avoid traditional 'guess-and-check' verification routines, we invert the standard governing stress and strain inequalities. By isolating geometric parameters (such as the standard module (m_{std}) and pitch diameters (d)) as dependent functions of material strength limits, we can calculate physical component mass directly from upfront preliminary specifications. Sizing calculations for every primary component flow step-by-step from the fundamental mass-volume relationship:

$$W = \rho \cdot V \quad (5.5)$$

5.4.1 Gears Mass

The total rotating gear mass is the sum of the material volumes of the spiral bevel pinion and gear, plus the planetary sun, planets, and outer ring gear shells multiplied by the gear steel density ρ_g :

$$W_{\text{gears}} = W_{\text{bevel}} + W_{\text{planetary gears}} \quad (5.6)$$

- **Spiral Bevel Stage Sizing:** Spiral bevel gears are modelled as conical disks. According to standard AGMA 2003 criteria, their outer pitch diameters must scale to satisfy surface contact fatigue (Hertzian pitting) thresholds under heavy combined thrust loads. The

minimum pinion outer pitch diameter:

$$d_{b, \text{pinion}} = \left(\frac{2 \cdot T_{\text{design bevel}}}{\lambda_b \cdot I_b \cdot \left(\frac{\sigma_{\text{allow, H, bevel}}}{Z_E} \right)^2} \right)^{1/3} \quad (5.7)$$

Integrating the conical volumes yields the combined mass envelope:

$$W_{\text{bevel}} = \left(\frac{\pi}{6} \cdot \rho_g \cdot C_w \cdot \lambda_b \right) \cdot (1 + u_b^3) \cdot (d_{b, \text{pinion}})^3 \quad (5.8)$$

Substituting the outer pitch diameter boundary profile directly into the volume calculation eliminates the face width ratio (λ_b), isolating the weight footprint strictly to raw torque and material parameters:

$$W_{\text{bevel}} = \left(\frac{\pi \cdot \rho_g \cdot C_w \cdot Z_E^2 \cdot (1 + u_b^3)}{3 \cdot I_b \cdot (\sigma_{\text{allow, H, bevel}})^2} \right) \cdot T_{\text{design bevel}} \quad (5.9)$$

- **Planetary Stage Sizing:** The planetary stage consists of 1 central Sun gear (s), N_p orbiting Planet gears (p), and 1 stationary internal Ring gear (r). The base volume of the gear blanks is defined by summing the individual cylindrical envelopes:

$$V_{\text{planetary}} = V_{\text{sun}} + N_p \cdot V_{\text{planets}} + V_{\text{ring}} \quad (5.10)$$

$$V_{\text{planetary}} = \pi \cdot \frac{d_s^2}{4} \cdot FC_w + N_p \cdot \left(\pi \cdot \frac{d_p^2}{4} \right) \cdot FC_w + \frac{\pi}{4} \cdot (d_{\text{outer, ring}}^2 - d_r^2) \cdot FC_w \quad (5.11)$$

Where F is the face width. To prevent severe radial strain from opening up the gear meshes as the planets pass by, the outer ring gear envelope is modelled as a realistic hollow shell with a protective wall thickness proportional to the tooth module:

$$d_{\text{outer, ring}} = d_r + 2C_t \cdot m_{\text{std}} \quad (5.12)$$

Expressing all diameters using the metric standard module ($d = m_{\text{std}} \cdot z$) and substituting the coaxial kinematic rules ($z_r = (u_p - 1)z_s$) and ($z_p = \frac{u_p - 2}{2}z_s$), for fixed ring gear configuration, the total planetary gear weight simplifies to:

$$W_{\text{planetary gears}} = \left(\frac{\pi}{4} \cdot \rho_g \cdot C_w \cdot \lambda_d \right) m_{\text{std}}^3 \cdot \left(z_s^3 \left(1 + N_p \left(\frac{u_p - 2}{2} \right)^2 \right) + (4C_t (u_p - 1) z_s^2 + 4C_t^2 z_s) \right) \quad (5.13)$$

To establish the boundaries, the required module must satisfy both AGMA root bending fatigue and contact pressure simultaneously:

$$m_{\text{bending}} = \left(\frac{2 \cdot T_{\text{design planet}}}{N_p \cdot z_s^2 \cdot \lambda_d \cdot \sigma_{\text{allow, F}} \cdot Y} \right)^{1/3} \quad (5.14)$$

$$m_{\text{contact}} = \left(\frac{2 \cdot z_E^2 \cdot T_{\text{design planet}}}{N_p \cdot z_s^3 \cdot \lambda_d \cdot I_p \cdot \sigma_{\text{allow, H, planet}}^2} \right)^{1/3} \quad (5.15)$$

Where Y is the Lewis form factor, and λ_d is the face-width-to-diameter ratio (F / d_s).

To ensure complete non-linear closure, the sun teeth parameter is locked to its absolute geometric floor to eliminate manufacturing tool undercutting:

$$z_s = \frac{2 \cos^3 \beta}{\sin^2 \alpha_n} \quad (5.16)$$

The constraint is determined via the maximum envelope function Equation 5.17 which is rounded up to the nearest discrete tool cutter size from the standard ISO-metric catalog ($m_{\text{std}} \geq m_{\text{calc}}$).

$$m_{\text{calc}} = \max(m_{\text{bending}}, m_{\text{contact}}) \quad (5.17)$$

Substituting the dominant contact-driven module expression into the cubic core completely eliminates the tooth count placeholders from the baseline diameter, yielding the substituted gear mass function:

$$W_{\text{planetary gears}} = \left(\frac{\pi \rho_g C_w T_{\text{design planet}}}{2 N_p I_p \sigma_{\text{allow, H, planet}}^2} \right) \cdot \left(\left(1 + N_p \left(\frac{u_p - 2}{2} \right)^2 \right) + \frac{4 C_t (u_p - 1)}{z_s} + \frac{4 C_t^2}{z_s^2} \right) \quad (5.18)$$

5.4.2 Structural Transmission Shafts and Pins Mass

The MGB shaft assembly weight tracks three active loading paths: the horizontal engine input shafts, the vertical transition sun shaft, and the planet pins.

Because the planet gears encircle the central sun axis symmetrically, their radial separation vectors push inward from opposite directions simultaneously, causing the net global bending moments on the main line components to cancel completely to zero. Thus, the sun and carrier shafts experience pure torsion (T).

Sized using the ASME transmission shaft design code, their safe diameters match the distortion energy (von Mises) fatigue criteria:

$$d_{\text{torsion}} = \left(\frac{16 \sqrt{3} \cdot n_s \cdot T}{\pi \cdot \sigma_{\text{allow shaft}}} \right)^{1/3} \quad (5.19)$$

The physical lengths of your shafts are strictly bound by the width of the gears mounted on them plus a fixed assembly clearance ($L_{\text{clearance}}$) for bearings and oil seals:

- Sun Shaft Length (L_s), must stretch across the gear face width:

$$L_s = F = \lambda_d \cdot m_{\text{std}} \cdot z_s \quad (5.20)$$

- Carrier Shaft Length (L_c), must clear the gear face width plus the thickness of the two structural carrier plates capping it:

$$L_c = F + 2 \cdot t_{\text{plate carrier}} = \lambda_d \cdot m_{\text{std}} \cdot z_s \cdot (1 + 2 \cdot C_c) \quad (5.21)$$

By substituting the gear-blank face width definition ($F = \lambda_d \cdot m_{std} \cdot z_s$) to define the physical shaft spans, we eliminate the independent length variables. The linear scaling weights for the input and output shafts are defined strictly by torque, tooth count, module, and material bounds:

$$W_{\text{sun shaft}} = \frac{\pi}{4} \rho_s \cdot \lambda_d \cdot m_{std} \cdot z_s \cdot \left(\frac{16 \sqrt{3} \cdot n_s \cdot T_{\text{design bevel}}}{\pi \cdot \sigma_{\text{allow shaft}}} \right)^{2/3} \quad (5.22)$$

$$W_{\text{sun shaft}} = \frac{\pi}{4} \rho_s \cdot \lambda_d \cdot m_{std} \cdot z_s \cdot (1 + 2 \cdot C_c) \cdot \left(\frac{16 \sqrt{3} \cdot n_s \cdot u_p \cdot T_{\text{design bevel}}}{\pi \cdot \sigma_{\text{allow shaft}}} \right)^{2/3} \quad (5.23)$$

Note: The only unknown left inside these bracket structures is the planetary stage gear ratio (u_p), as the module (m_{std}), and sun teeth (z_s) are resolved beforehand.

Conversely, the individual planet pins are not balanced. Each pin acts as a short, localized beam of length F supporting a heavy mechanical bending load. The net radial force concentrated on each pin is:

$$W_{\text{pin}} = \frac{4T_{\text{design planet}}}{m_{std} \cdot z_s \cdot N_p} \quad (5.24)$$

Treating the pin as supported on both ends by the rigid carrier walls, the peak internal bending moment occurs at the center of the pin span ($M_{\text{pin}} = \frac{W_{\text{pin}} \cdot F}{4}$) simplifies directly to:

$$M_{\text{pin}} = \frac{\lambda_d \cdot T_{\text{design planet}}}{N_p} \quad (5.25)$$

Sized under pure bending fatigue ($T = 0$), the total pin mass across all (N_p) paths scales as:

$$W_{\text{pins}} = \frac{\pi}{4} \cdot \rho_s \cdot N_p \cdot \lambda_d \cdot m_{std} \cdot z_s \cdot \left(\frac{32 \cdot n_s \cdot \left(\frac{\lambda_d \cdot T_{\text{design planet}}}{N_p} \right)}{\pi \cdot \sigma_{\text{allow shaft}}} \right)^{2/3} \quad (5.26)$$

5.4.3 Rigid Planet Carrier Mass

The planet carrier is a structural frame flanking the gears that collects the planet pin loads to rotate the main rotor mast. To protect the gear meshes from edge-loading stress concentrations, it must be highly rigid against torsional twist strain. Sized as twin circular plates of diameter matching the outer ring boundary ($d_{\text{outer ring}}$) and plate thicknesses proportional to the gear width ($t_{\text{plate carrier}} = C_c \cdot F$), its structural mass is derived directly from its geometric envelope:

$$W_{\text{carrier}} = 2 \left(\frac{\pi}{4} d_{\text{outer ring}}^2 (C_c \cdot F) \right) \cdot \rho_s \cdot K_{\text{web}} \quad (5.27)$$

Where K_{web} is the geometric modifier accounting for weight-reduction pockets pocketed out of the carrier web plates $0.60 \leq K_{\text{web}} \leq 0.75$.

By substituting the gear-blank face width and pitch diameter parameters under the assumption that contact stress governs the active sizing envelope, the carrier mass equation simplifies to:

$$W_{\text{planetary gears}} = \left(\frac{\pi \cdot \rho_s \cdot C_c \cdot K_{\text{web}} \cdot Z_E^2 \cdot (u_p - 1)^2}{N_p \cdot I_p \cdot \sigma_{\text{allow, H, planet}}^2} \right) \cdot T_{\text{design planet}} \quad (5.28)$$

5.4.4 Bearings Mass

Sized using an analytical load-scaling law calibrated from aerospace bearing catalogs, the mass of rolling elements scales with the net localized radial force vector raised to the 1.2 power. Expressing the radial forces on the planet pins and the main lines directly via torque, speed, and standard module variables results in the combined bearing mass expression:

$$W_{\text{bearings}} = N_p \cdot \left(X_{\text{planet}} \cdot \left(\frac{4 \cdot T_{\text{design planet}}}{m_{\text{std}} \cdot z_s \cdot N_p} \right)^{1.2} \right) + X_{\text{shaft}} \cdot \left(u_{\text{overall}} \cdot T_{\text{design bevel}} \right)^{0.8} \quad (5.29)$$

Where X_{planet} and X_{shaft} represent empirical lookup constants calibrated relative to the primary shaft dimensions.

5.4.5 Structural Housing Casing Mass

The casing forms the environmental boundary layer, structural load path, and pressure shell enclosing both the spiral bevel input meshes and the final planetary stage. To minimize structural dead-weight, the housing is modeled as a unified, tapered Conical Tower Shell rather than a uniform cylinder, matching the tapering force profile of the internally stacked stages.

Sized using an ultra-lightweight magnesium alloy, the total housing mass is calculated as the sum of the upper planetary cylinder barrel, the lower tapered bevel cone frustum shell, and the structural sealing end plates:

$$\begin{aligned} W_{\text{housing}} = \rho_h \cdot \pi \cdot K_{\text{rib}} \cdot & \left((D_{h, \text{planet}} \cdot t_{\text{case}} \cdot L_{h, \text{planet}}) \right. \\ & + \left(t_{\text{case}} \cdot L_{h, \text{bevel}} \cdot \frac{D_{h, \text{planet}} + d_{b, \text{pinion}}}{2} \right) \\ & \left. + \left(\frac{t_{\text{plate}}}{2} \cdot D_{h, \text{planet}}^2 \right) \right) \end{aligned} \quad (5.30)$$

Where the active boundary diameter and length profiles are mapped strictly from your standard module and integer tooth boundaries:

- Planetary Upper Housing Diameter:

$$D_{h, \text{planet}} = (m_{\text{std}} \cdot z_s + 2 C_t \cdot m_{\text{std}}) + 2 t_{\text{case}} \quad (5.31)$$

- Planetary Stage Housing Length:

$$L_{h, \text{planet}} = \lambda_d \cdot m_{\text{std}} \cdot z_s \cdot (1 + 2 C_c) + L_{\text{clearance}} \quad (5.32)$$

- Bevel Stage Housing Length:

$$L_{h, \text{bevel}} = d_{b, \text{pinion}} \cdot \tan(\delta_{\text{gear}}) + 2 \cdot L_{\text{clearance}} \quad (5.33)$$

The factor ($K_{\text{rib}} \approx 1.20$) is the structural multiplier that adds the physical weight of the deep external grid-stiffness ribs cast onto both the cylinder and conical perimeter walls to prevent casing deflection under flight loads.

5.5 Global Multi-Stage Ratio Optimization Split

The global multi-stage ratio optimization ensures minimum system mass by deriving a weight minimization penalty function, focusing on the bevel ratio (u_b) and planetary ratio (u_p). By expanding the derivative of this system function with respect to u_b , a master optimization of quintic polynomial is established.

From our previous derivations, we proved that the physical mass of a gear blank scales directly with its torque capacity, which is controlled by its gear ratio.

1. The Spiral Bevel Stage Weight: Scales cubically with its own ratio because it is a solid conical wheel setup: $W_b \propto u_b^3$.
2. The Planetary Stage Ring Gear Weight: The outer ring gear envelope dominates the final planetary weight. Its diameter scales linearly with its ratio, and its volume scales cubically with that diameter: $W_{\text{ring}} \propto (u_p - 1)^3$.

Because the planetary stage is downstream, it receives the torque multiplied by the first stage. Therefore, we must multiply the planetary weight by the input stage ratio (u_b) to account for the torque scale-up: $W_{\text{planetary scaled}} \propto u_b \cdot (u_p - 1)^3$.

3. Shaft Assembly Mass:

- Bevel Shafts: $\propto u_b^{2/3}$
- Planetary Shafts: Constant relative to the split
- Pins: $W_{\text{pins}} \propto u_b^{4/3}$

4. Planet Carrier Mass: $\propto u_b \cdot \left(\frac{u_{\text{overall}}}{u_b} - 1\right)^2$

5. Bearings Mass: Constant relative to the split

6. Housing Casing Mass: $\propto \frac{u_{\text{overall}}}{u_b}$

If we add the structural scaling metrics of the housing, carrier, and shafts to our baseline gear blank equations, the comprehensive mass penalty function $f(u_b)$ expands to:

$$\begin{aligned}
 f(u_b) = & (1 + u_b^3) \\
 & + u_b \left(\frac{u_{\text{overall}}}{u_b} - 1 \right)^3 \\
 & + \gamma_1 u_b^{\frac{2}{3}} + \gamma_2 u_b \left(\frac{u_{\text{overall}}}{u_b} - 1 \right)^2 \\
 & + \gamma_3 \left(\frac{u_{\text{overall}}}{u_b} \right) + \gamma_4 u_b^{\frac{4}{3}}
 \end{aligned} \tag{5.34}$$

Where γ_1 , γ_2 , γ_3 , and γ_4 are structural mass distribution constants determined by your material choices and geometric properties.

5.6 Analytical Minimization and Derivation of the Master Quintic Polynomial

To resolve the precise architectural configuration that yields the absolute minimum mass for the entire assembled Main Gearbox (MGB), calculus must be applied to the comprehensive system function.

To locate the absolute minimum mass floor, we evaluate the first derivative of the expanded system mass function with respect to our primary design variable (u_b), setting the resulting expression equal to zero: $\frac{\partial f}{\partial u_b} = 0$.

Once your optimization code calculates u_b , the engineering sequence unlocks completely.

The software can now run down the line to calculate the exact tooth sizes, shaft dimensions, and material volumes for every single component in your transmission tower. A final linear summation of all five computed sub-assembly blocks to output the exact flight weight of the finished transmission assembly.

5.7 Comprehensive Housing Size

To calculate the spatial bounding box of the complete, fully enclosed helicopter main gearbox tower, we translate our discrete design coordinates (m_{std} , z_s , z_r , and $d_{b, \text{pinion}}$) straight into a physical three-dimensional envelope. Because the architecture wraps both the narrower horizontal spiral bevel input and the wider vertical planetary cell into a single, cohesive structural shell, the geometric package is completely defined by two primary parameters: the Maximum System Outer Housing Diameter ($d_{h, \text{planet}}$) and the Total Assembly Tower Height (H_{sys}).

5.7.1 The Maximum System Outer Diameter

The maximum width across the circular perimeter profile of the transmission occurs at the upper cylindrical section flanking the stationary planetary ring gear:

$$D_{h, \text{planet}} = m_{std} \cdot z_s \cdot \left((u_p - 1) + \frac{2C_t}{z_s} \right) + 2t_{case} \quad (5.35)$$

5.7.2 The Total Housing Tower Height

The total axial length of the gearbox shell represents the direct stacking of the vertical planetary stage thickness, the spiral bevel cone drop length, and the required internal bearing arrangement clearances.

$$H_{sys} = L_{h, \text{planet}} + L_{h, \text{bevel}} \quad (5.36)$$

$$H_{sys} = \left(\lambda_d \cdot m_{std} \cdot z_s \cdot (1 + 2 \cdot C_c) + L_{clearance} \right) + \left(d_{b, \text{pinion}} \cdot \tan \delta_{gear} + 2 \cdot L_{clearance} \right) \quad (5.37)$$

5.7.3 Total Volumetric Size

By treating the casing structure as an upper cylinder intersecting a lower tapered cone frustum shell, the literal volumetric space occupied is:

$$V_{\text{envelope}} = \frac{\pi}{4} \cdot D_{\text{h, planet}}^2 \cdot L_{\text{h, planet}} + \frac{\pi}{12} \cdot L_{\text{h, bevel}} \cdot \left(D_{\text{h, planet}}^2 + D_{\text{h, planet}} \cdot d_{\text{b, pinion}} + d_{\text{b, pinion}}^2 \right) \quad (5.38)$$

Chapter 6: Romax

6.1 Introduction

Romax Technology supplies comprehensive simulation software for designing, analyzing, and optimizing drivetrains and rotating machinery across automotive, wind energy, aerospace, and marine sectors. Its tool suite covers durability testing, gearbox and bearing design, structural analysis, and early stage performance rating, enabling rapid, intuitive modelling that improves efficiency, reliability and overall mechanical performance. Romax is recognized as a trusted partner for end-to-end drivetrain development, providing specialized capabilities for electric machine structural design and full powertrain optimization.

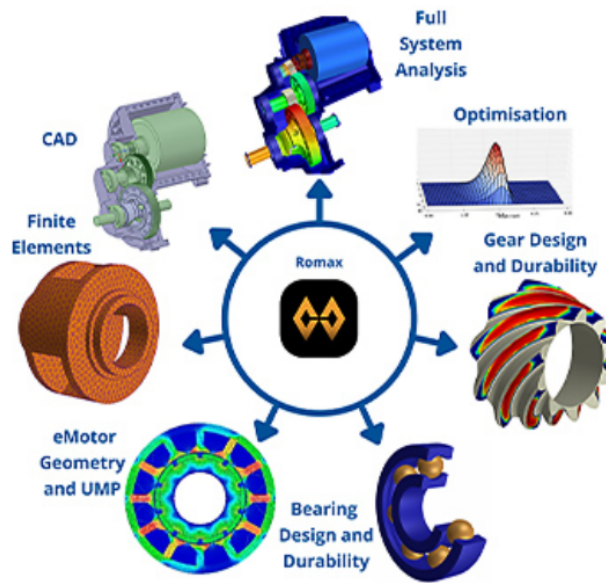


Figure 6.1: Romax

6.2 Software Definition

The Romax Software Suite is a physics-based Computer-Aided Engineering (CAE) platform built for modeling, simulating, and optimizing rotating machinery and electrified powertrains. Moving away from traditional isolated component design, Romax uses a holistic system architecture. This methodology captures the cumulative, non-linear physical interactions across an entire transmission system under dynamic operational loads, considering the simultaneous effects of component deflections, gear misalignments, shifting contacts, housing flexibility, and non-linear bearing stiffness.

6.3 The Romax Hybrid System-Simulation Approach

To accurately analyze these complex interactions, Romax software uses a hybrid modeling approach that integrates multiple computational methods into a single, coupled simulation, balancing calculation speed with high prediction accuracy:

- **Finite Element (FE) Method:** Condenses large structures, such as flexible gear housings or planet carriers, into reduced substructures to capture structural stiffness and mass distribution efficiently.
- **Analytical Formulations:** Computes shaft bending lines and gear meshing mechanics using closed-form analytical equations and advanced gear/bearing contact models.
- **Empirical Formulations:** Estimates bearing clearances and operational fatigue lives according to standard codes, such as international rolling bearing rating life methods.

6.4 Features and Key Benefits

Romax offers a comprehensive CAE solution for drivetrain and powertrain engineering, delivering cutting-edge simulation and analysis capabilities that empower engineers to design, optimize, and validate complex systems with precision and speed.

The product delivers a holistic system view, capturing intricate component interactions and enabling design of the entire system from physics based, parametric definitions for every part; it relies on proven, proprietary technology that intelligently blends analytical, empirical, and finite element approaches, coupled with world-class gear and bearing contact models. By allowing models to be built and evaluated in minutes, it delivers speed and simplicity, providing rapid, actionable results that let engineers run “what if” scenarios and gain instant insight. It encourages CAE-driven design by merging easily editable models with repeatable, multi-fidelity analyses and multi-attribute optimisation, empowering better engineering decisions and reducing innovation risk. The flexible product suite streamlines processes with collaborative workflows, optimization, and automation that boost productivity. Integrated into the broader Hexagon ecosystem, the tool breaks organizational barriers through seamless connections between CAE, CAD, and FEA, specifically enhancing powertrain design and beyond.

6.5 Integrated Applications

The platform architecture utilizes six distinct, interoperable modules engineered to support collaborative engineering workflows across cross-functional powertrain development teams:

- **Romax Enduro:** The primary structural durability interface used to perform fatigue evaluations, component structural safety factor checks, and system deflection analysis.
- **Romax Spectrum:** A highly specialized vibration and acoustics module designed to model dynamic transmission error (TE), predict structural noise paths, and map housing acoustic radiation. Also, it facilitates the simulation of noise and vibration (NVH) in electro-mechanical powertrains.

- **Romax Spin:** An advanced internal element modeling workspace allowing bearing specialists to map roller-edge stress concentrations, local contact footings, and internal contact geometry.
- **Romax Concept:** A streamlined workspace optimized for clean-sheet structural layout definition and spatial envelope exploration. Also, it can help investigate EV drivetrain concepts in their early stages.
- **Romax Energy:** A dedicated efficiency and fluid-loss solver that isolates and quantifies gear mesh friction, rolling element resistance, and hydrodynamic oil churning losses.
- **Romax Evolve:** An electro-mechanical simulation module that couples stator and rotor structural dynamics with multi-phase electromagnetic forces in unified electric vehicle powertrains. Also, it allows for structural design and analysis of electrical machines.

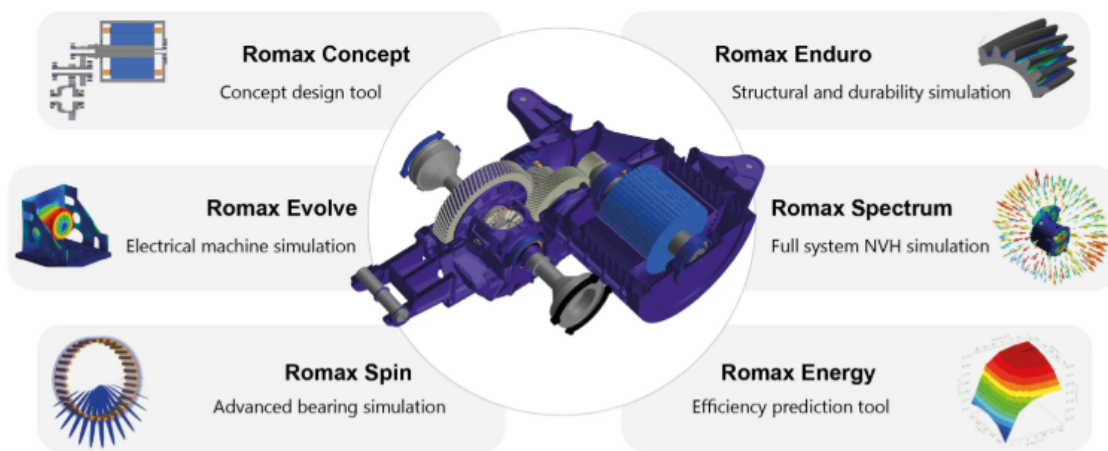


Figure 6.2: Romax

6.6 Automated System Sizing Interfaces

To transition a preliminary generative design into a validated powertrain, the structural boundaries established by the closed-form weight and sizing equations are translated directly into the physics-based environment of the Romax Concept. Rather than operating as an independent, isolated component model, the software uses a multi-layered parametric data pipeline.

6.6.1 Romax Primary System Inputs

The analytical model requires a rigorous data matrix defining the full macro-geometric, kinematic, and operational layout of the transmission before it can execute any non-linear structural sweeps:

- **Operational Conditions and Duty Cycles:** The software requires explicit multi-step loading profiles defining nominal input power, engine input velocity, and target main rotor mast velocity.
- **Gear Macro-Geometry:** The system architecture requires the exact parameters derived from the optimization loop: standard metric module m_{sta} , pressure angle α_n , helix angle

β , and specialized spiral bevel tooth indices. Sizing profiles must contain explicit teeth counts—the sun teeth z_s , planet teeth z_p , and ring teeth z_r —along with the precise face widths F_b and F_p .

- **Planetary Gearbox Boundaries:** The spatial inputs include the precise number of active planet gears and the geometric layout positions of the planet pins relative to the vertical centre line.
- **Shaft Geometry:** Shaft inputs discard one-dimensional line assumptions, requiring complete 2D/3D profile coordinate matrices. The pipeline defines explicit shaft length segments along with outer shaft step diameters, internal hollowing parameters for weight reduction, and the physical dimensions of the planet pins.
- **Rolling Element Bearing Selection:** Sizing requires either direct lookup selection from integrated commercial manufacturer databases (e.g., SKF, NSK, Timken catalog lines) or explicit definition of internal rolling geometry. This includes roller/ball counts, raceway diameters, pitch circles, internal clearances, and the specific structural preload vectors acting on the input and output main line supports.
- **Material Fatigue and Lubricant Limits:** Sizing require exact core and case material properties, including densities, Young’s Modulus, Poisson’s ratio, and specific allowable fatigue limits. Fluid mechanics constraints require the lubricant type, dynamic viscosity indices, operating temperatures, and oil-jet delivery pressures.

6.6.2 Romax Multi-Disciplinary Outputs

Once the parametric input is compiled, the solvers execute a whole-system structural study. The software returns four primary engineering evaluation modules:

- **Stress Analysis and Component Safety Factors:** The system outputs exact, code-validated safety margins. It logs individual safety factors for tooth root bending, surface contact pressure, and shaft torsional shear fatigue.
- **Whole-System Mass Bookkeeping and Spatial Size:** Romax treats all gears, shafts, and casing components as true 3D solid geometries. It integrates the exact volume of the thin-walled housing and the webbed gear wheels to output the Total Assembled Weight.

6.7 Romax in AI/ML Based Transmission Design

The paper shows how Romax’s physics based; full system gearbox simulation platform can be combined with ODYSSEE’s AI/ML tools to create a practical, high-performance optimization loop. It uses AI to accelerate and guide the traditional physics-based simulation workflow.

It isn’t exactly what the project will do, but it gives an idea of how Romax will be used and how AI improves traditional calculations. The concept is like the GAIT project: instead of calculating gearbox weight and size one by one with many formulas, AI assists and replaces traditional hand calculations. With Romax, we can also verify the results.

It outlines a shift in transmission design driven by sustainability, electrification, and shorter development cycles. While conventional CAE is mature, AI/ML can accelerate the simulation of lifecycle, democratize advanced engineering, and reduce risk. The authors fuse physics-based CAE with AI/ML, moving from simulation test validation to Design-of-Experiments fed

machine learning models verified by simulation and experiment. Two case studies illustrate this: a gear-micro-geometry optimization that reduces transmission error, and a larger study that iteratively optimizes numerous design parameters against broad performance criteria. The method applies beyond automotive—into aerospace and other sectors—enabling full design space exploration, stronger multiphysics optimization, and ensuring gearboxes match simulated behavior. The ultimate aim is faster, higher quality powertrain development with less reliance on specialist expertise.

- **Romax**

Parametric, hybrid FEM or empirical or analytical simulation of entire gearboxes (gears, shafts, bearings, CVT, etc.). Fast (<1s per run) and highly accurate for gearbox mechanics, durability, NVH, efficiency, weight plus packaging.

- **ODYSEE**

AI / ML platform that builds Reduced-Order Models (ROMs) from a handful of selected simulation runs and then performs design-of-experiments (DoE), sensitivity analysis, and optimisation on that ROM.

- **Romax to ODYSSEE to Romax**

1. Define the full system model in Romax (parametric).
2. Use Romax batch-run to generate a small training data set.
3. Train a ROM in ODYSSEE.
4. Run millions of parametric evaluations on the ROM (but actually far fewer than would be needed in Romax).
5. Identify the best parameter combinations.
6. Re-run the selected candidates in full Romax to validate and finalize the design.

Chapter 7: Transitioning from Traditional Optimization to AI or ML Models

7.1 The Classical Paradigm: Simulation Validated by Physical Test

Historically, the powertrain optimization lifecycle has operated as a linear, reactive, and highly deterministic loop. This methodology follows a rigid progression:

1. **Empiric Sizing:** An engineer initiates a baseline configuration using classical closed-form mechanics and design rules.
2. **Deterministic Simulation:** The baseline model is subjected to finite element and analytical systems simulation to check for macro-structural deflections, stress concentrations, and systemic misalignments.
3. **Empirical Validation:** A physical metal prototype is manufactured following specific material treatments and subjected to endurance testing on a dynamometer rig to establish the ground truth.
4. **Reactive Calibration:** Disagreements between physical test failures and predictions require the engineer to retroactively calibrate simulation boundary conditions, modify the CAD geometry, and repeat the cycle.

7.1.1 Systemic Constraints of the Classical Paradigm

While effective for simple systems, these traditional optimization workflows hit strict scalability limits when applied to high-fidelity powertrain systems—particularly when evaluating robustness against manufacturing variations across multiple stages simultaneously.

Because physical prototyping requires long lead times and significant capital expenditure, engineering teams can realistically evaluate only two or three macro-design iterations. Consequently, the final architecture is rarely a true global optimum; it is simply the first functional variation that satisfies minimum safety margins.

7.2 The Modern Frontier: DoE-Fed AI Models Validated by Simulation and Test

To accelerate this design cycle, the industry is transitioning to an AI/ML-driven workflow. The integration of high-velocity automation scripts (such as the Romax Python API) and machine

learning transforms the traditional design lifecycle. It replaces a reactive, human-driven loop with an automated, predictive optimization framework.

This modern methodology maps to a four-phase execution pipeline:

- **Phase 1: Automated DoE Generation**

Rather than modifying designs one variable at a time, engineers leverage programmatic execution scripts to establish a massive Design of Experiments (DoE) matrix. The script automatically varies thousands of coupled geometric parameters—such as housing wall topology, gear face widths, bearing positions, and macro-micro tooth profiles—across a wide mathematical spectrum.

- **Phase 2: High-Velocity Simulation Processing**

This automated DoE matrix is pushed directly to high-fidelity parallelized simulation engines. The system computes the precise weight, structural stiffness, safety factors, and noise, vibration, and harshness (NVH) characteristics for every variation. This brute-force computation converts raw geometric permutations into a comprehensive, high-fidelity engineering dataset.

- **Phase 3: Deep Learning and Surrogate Modeling**

This compiled dataset is used to train a machine learning algorithm, creating an advanced **AI Model**. The AI maps the complex, non-linear relationships across variables, learning instantly how a micro-geometric adjustment to a gear tooth profile impacts global housing stiffness, system efficiency, and dynamic transmission error. Once trained, these predictive models run orders of magnitude faster than full numerical simulations, predicting the performance of new configurations in milliseconds and enabling real-time exploration of the design space.

- **Phase 4: Dual-Gate Validation**

To ensure structural safety and verify design robustness before committing to physical manufacturing, the predictive outputs of the AI are verified through a targeted dual-gate validation process. First, high-fidelity system simulation is used to double-check the top optimized candidates flagged by the AI. Second, a single physical prototype is built to confirm the final accuracy of the model, transforming the physical test bench from an iterative troubleshooting platform into a final quality assurance test.

7.3 Industrial Implications of the Paradigm Shift

Flipping the engineering workflow from traditional simulation to AI-driven optimization delivers three major competitive advantages:

- **Reduction in Computational Lifecycle**

Complex multi-variable optimization loops that historically required weeks of engineering hours and continuous CAD adjustments are resolved by trained surrogate models in fractions of a second (moving from days to milliseconds).

- **Discovery of Unintuitive Solution Spaces**

Human designers naturally favor conservative, familiar design patterns. AI algorithms evaluate the entire mathematical design envelope without bias, discovering lightweight, high-efficiency topologies that standard workflows would overlook.

- **Decoupling of Development Costs**

Minimizing physical iteration loops reduces prototyping expenses. Instead of building five rounds of physical prototypes, engineering teams aggressively de-risk innovative designs in a virtual environment before committing capital to a single final physical manufacturing unit.

Chapter 8: Project Development

8.1 Introduction

Gearbox layout generation is the process of determining the number of reduction stages, shaft locations, gear ratios per stage, housing dimensions, gear arrangement and positioning, and bearing placement. The objectives of this process include meeting speed-reduction requirements, maximizing efficiency, respecting packaging constraints, transmitting torque safely, and minimizing weight.

- **Given:**

$P, n_{in}, n_{out}, T_{req}$, Material specs Operation condition, Material, Geometric input

- **Determine:**

$N_{stages}, u_i, z_{in}, m_i, a_i, b_i, D_{shaft}$

- **Subject to:**

$g(x) \leq 0$

- **Where constraints include:**

Fatigue Limit and Stress Limit

- **Where constraints include:**

Layout generation is not a sizing problem and it is a coupled topology-selection problem.

Traditional gearbox design philosophy relies on engineering handbooks, designer experience, design standards (AGMA, ISO), and previous gearbox templates. The approach is characterized by proven methodology and easy validation, but it is also time-consuming, requires expert knowledge, and offers limited design exploration. So, the value of this project is to minimize those constraints—reducing time consumption, lessening the need for specialized expertise, and expanding design exploration—by developing an AI system as a proof-of-concept that will aid engineers in estimating the optimal weight and size of the gearbox layout.

The sizing of the helicopter main gearbox follows a reference architecture in which Stage 1 consists of a spiral-bevel reduction that receives power from the engines and performs a (90°) axis redirection, while Stage 2 incorporates a five-planet planetary reduction that drives the rotor-mast epicyclic stage and a fixed ring gear.

8.2 Cohort Pipeline

The pipeline for this cohort is stated in the image below. The process begins with Python input, comprising operational condition, material, and geometric data. This information is

incorporated into a Python script, which generates outputs that become the parameters for Romax: geometry and location, number of teeth, reduction ratio, and face width. Romax then verifies whether these parameters satisfy weight and sizing constraints and whether the design achieves the required safety factor. The Romax output includes mass, stress analysis, the actual gear ratio, and the safety factor. If the design is valid, the team proceeds to AI model input; if not, the team updates and reconfigures the design script and re-runs the pipeline.

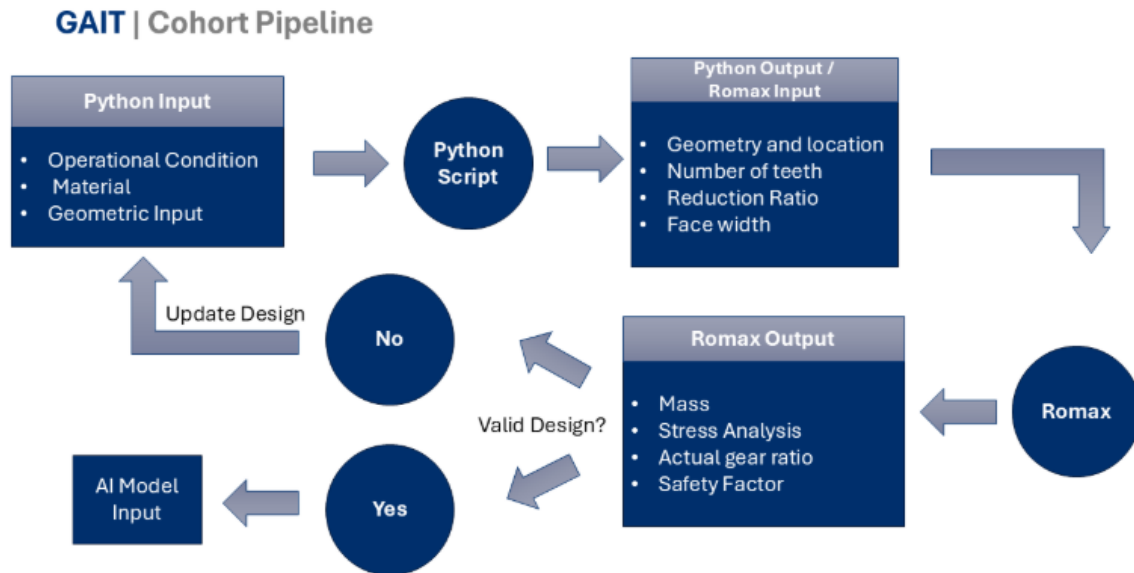


Figure 8.1: Cohort Pipeline

Following the cohort pipeline, the image below displays a screenshot from the Python script that produced the output parameters and sizing. The team then used these parameters to sketch and manually design the gearbox in Romax.

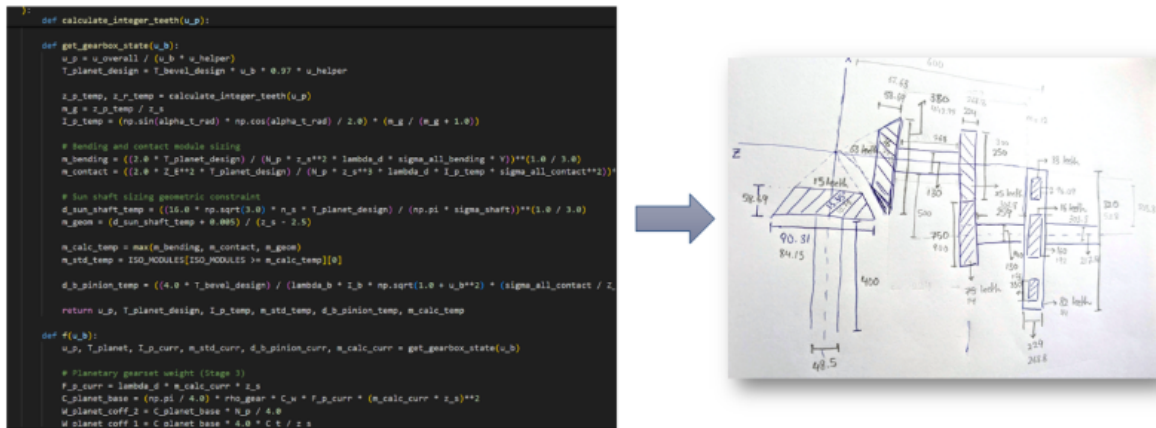


Figure 8.2: Python output to Romax input

Then, the Python output was used as input to Romax to generate the gearbox design.

8.3 Romax Concept (Student Edition)

Romax is the software used as a validation tool. During this time, the team used Romax Concept (Student Edition), the only Romax product that offers a student edition—a trial

version of the product. This helped the team gain an understanding of Romax in general and facilitated progress on the project.

Romax Concept (Student Edition) is a powerful tool for students and educators in the field of machine design and power transmission systems. It offers a comprehensive suite of tools for designing power transmissions, including systems of gears, shafts, and rolling-element bearings.

Take note that this version has limitations in features. The following features of Romax Concept are not available in the student edition:

- Importing or exporting of REXS model files into Concept
- The exporting of the Concept model to Romax DT model files
- The New from Template and Save as Template option
- The Detailed Design option of the Gear Set editor
- No access to Helical Gear Design/Optimisation tools

This version can specify, edit and model Gears using a Loading Design or Concept Design only. Also, it can load designs created with Romax Concept Student Edition only.

The Figure 8.3 shows the gearbox designs created as practice while the team was exploring the software.

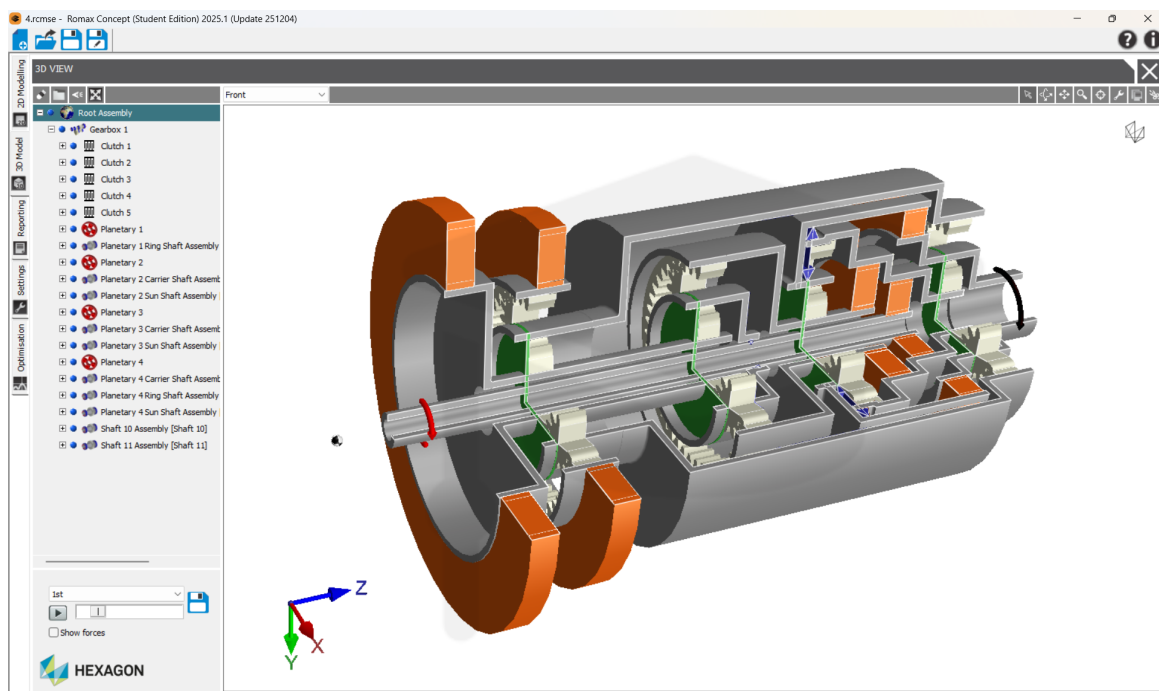


Figure 8.3: Romax Concept (Student Edition) Practice

Following the cohort pipeline, the parameters that were given by Python were input into Romax. The following are the Romax input parameters, and the image below shows the window where the data were input.

- Face width (mm)
- Number of teeth
- Pressure angle

- Helix angle
- Number of planets
- Material
- Desired gear ratio
- Mean pitch diameter (mm)
- Duration (hrs)
- Speed 1 (RPM)
- Power 1 (kW)

The screenshot displays the Romax Concept (Student Edition) software interface, divided into two main panels: 'Planetary' and 'Bevel Gear Set'.

Planetary Panel:

- Design Mode:** Loading Design, Concept Design, Detailed Design (Commercial version only).
- Configuration:**
 - ☒ Module / Tooth Numbers
 - ☒ Ratio and secondary gear teeth
 - ☐ Tooth numbers
 - ☐ Ratio and Module
- Primary gear:** Annulus gear
- Secondary gear:** Sun gear
- Ratio definition:**
 - Primary gear: Desired gear ratio: -2.000
 - Secondary gear: Actual gear ratio: -2.0
- Normal module:** 3.41000 mm
- ☐ Allow uneven planet spacing (if necessary)
- Table:**

Name	Teeth
Sun gear	22
Planet gear	11
Annulus gear	44
- Buttons:** NVH Assessment...

Bevel Gear Set Panel:

- Design Mode:** Loading Design, Concept Design
- Gear set name:** Bevel Gear Set 1
- Description:** (empty text area)
- Configuration:**
 - ☐ Straight bevel
 - ☐ Spiral bevel
 - ☒ Hypoid
- Pinion pressure angle (concave side):** 10.318 deg
- Pinion pressure angle (convex side):** 29.682 deg
- Pitch angle calculation for:** Large cutter
- Pinion mean spiral angle:** 54.378 deg
- Gear mean spiral angle:** 28.348 deg
- Gear ratio:** 3
- Pinion hand:** Left (Right button is highlighted)
- Pinion:** Gear 1
- Table:**

Name	Face width (mm)	Mean pitch diameter (mm)	User defined
Gear 1	30.000	96.509	✓
Gear 2	20.000	191.608	✓
- Table:**

Name	Normal backlash: (mm)	Coefficient of restitution:
Gear 1 -> Gear 2	0	0.7
- Buttons:** Material..., Cost...

Figure 8.4: Romax Concept (Student Edition) Input

After, the Romax gave the following output parameters:

- Actual gear ratio
- Safety Factor
- Contact fatigue
- Bending fatigue
- Torque 1 (Nm)
- Speed 2 (RPM)

- Torque 2 (Nm)
- Power 2 (kW)

The Figure 8.5 shows the window where the output was displayed. While it was summarized in the screenshot, the actual output includes a design summary and rating results that contain a detailed report of the gears.

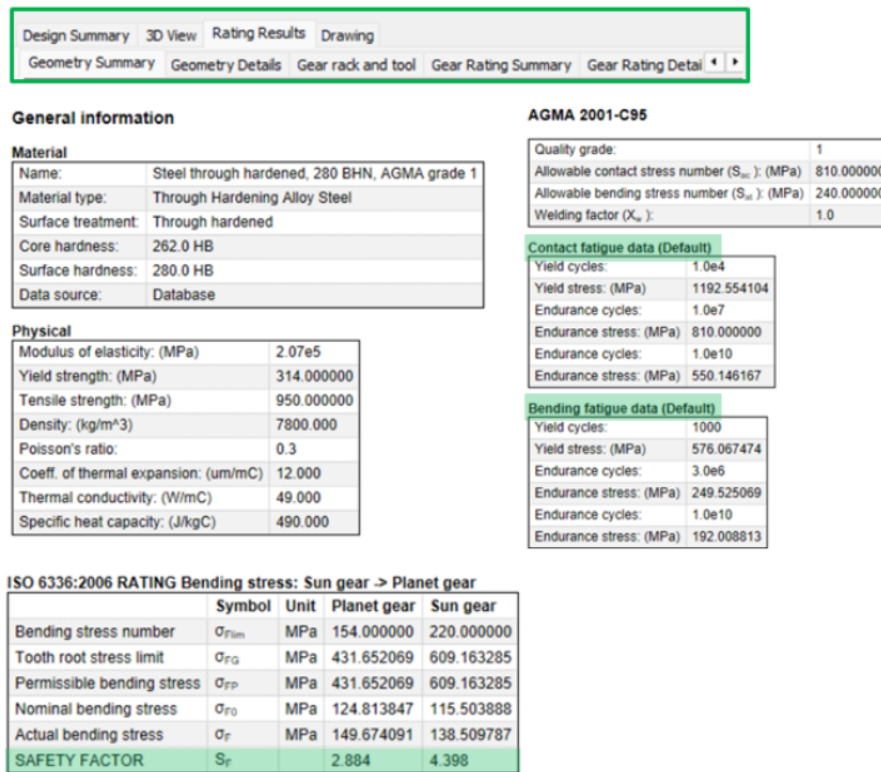


Figure 8.5: Romax Concept (Student Edition) Output

As part of the deliverables, the team needs to validate the gearbox weight and sizing layout. The image below shows the output, which includes Mass (kg) and size. The results are located under the Reporting tab in the Basic Geometry section of Romax and can also be exported to an Excel format. The image displays the following parameters: component, length (mm), mass (kg), volume (cc), and material for the shafts, and component, face width (mm), diameter (mm), mass (kg), volume (cc), material, and number of teeth for the gears.

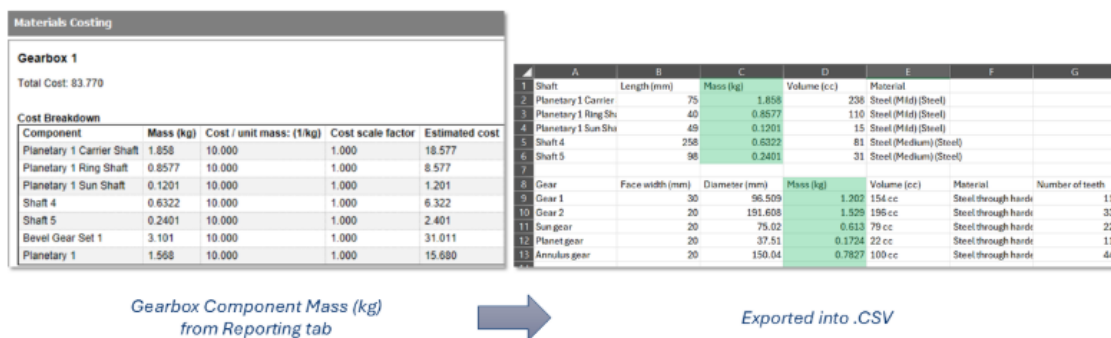


Figure 8.6: Gearbox weight and sizing

During the first validation, the design rating result from Romax indicated design failure along with a safety-factor failure; consequently, the team updated and reconfigured the Python script to address the identified issues.

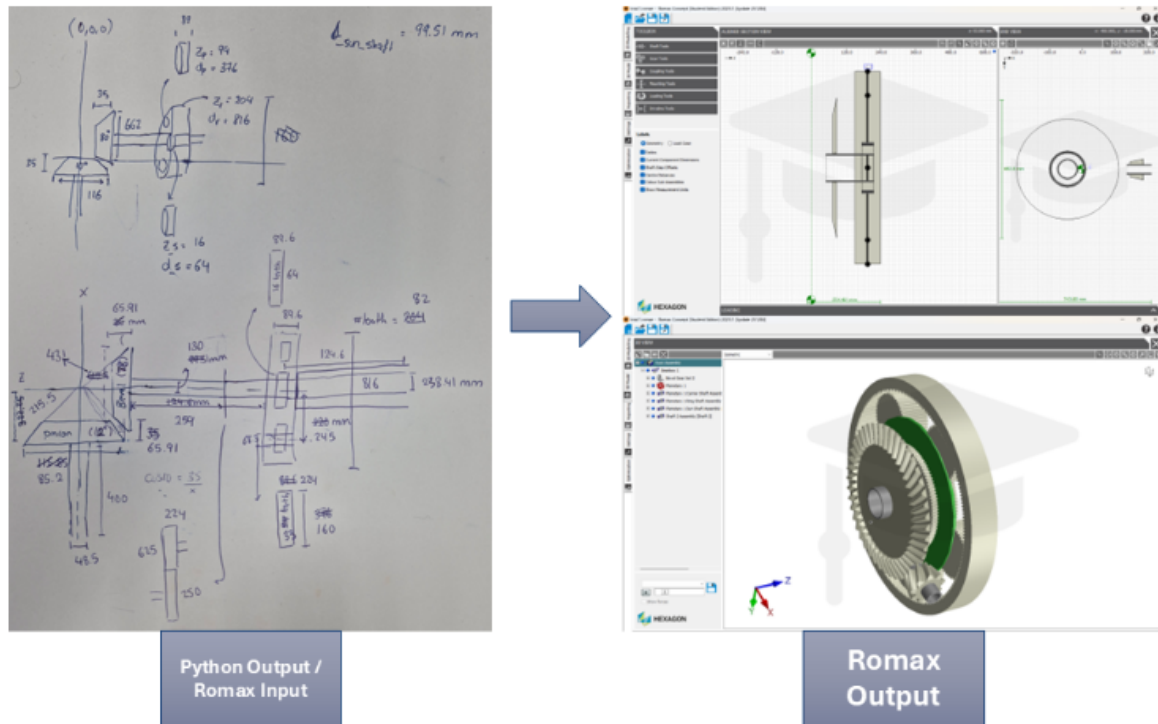


Figure 8.7: Gearbox Design Validation 1

After reconfiguring the Python script and following the cohort pipeline, Python gave the team a second design to be validated. This time, the gearbox weight (1,038 kg) appears close to the weight result in Python, which is 1,300 kg, considering that the Python result includes the gearbox casing and bearing weight. The design rating result also indicates success, as does the safety factor.

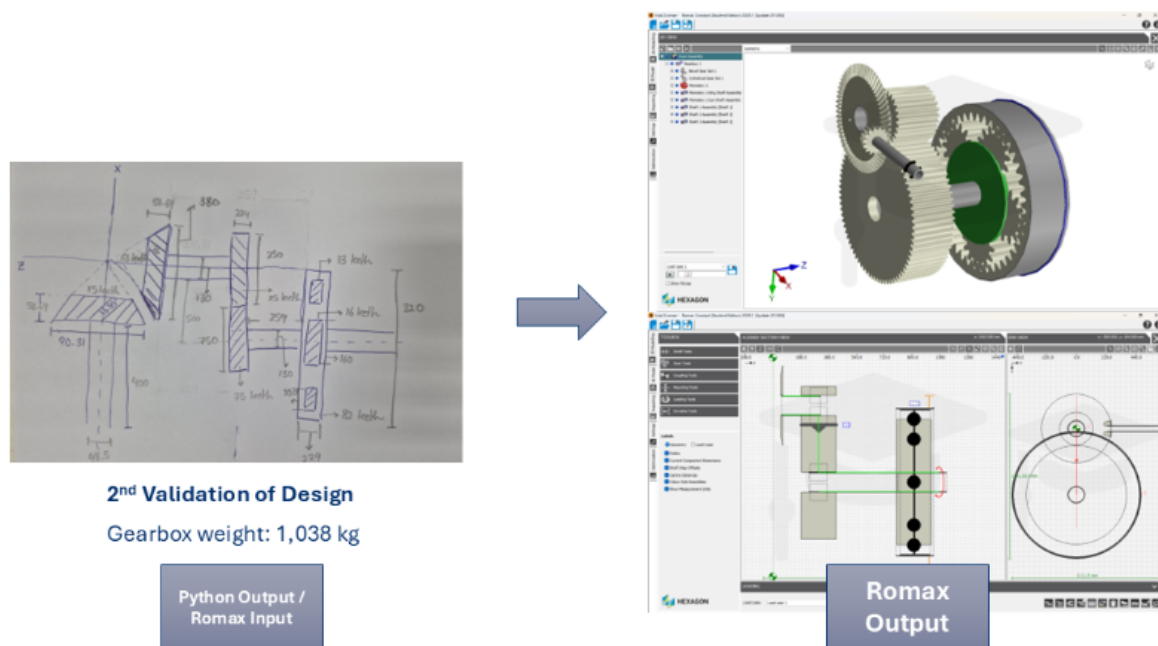


Figure 8.8: Gearbox Design Validation 2

After reviewing their initial results, the team redesigned the gearbox, basing the new arrangement on the A.R.T. paper that details split-torque gearbox types. They made the change because the older configuration could not efficiently transmit large amounts of power required.

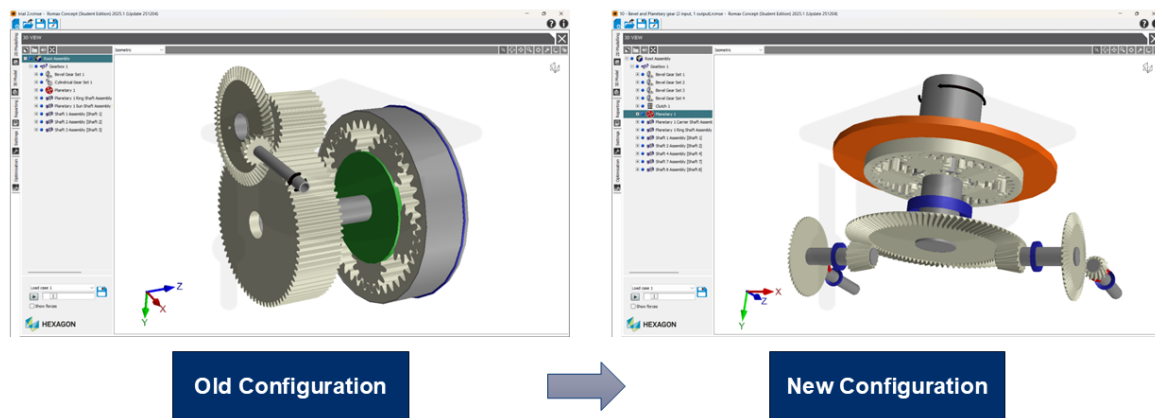


Figure 8.9: Gearbox Design New Configuration

In this new configuration, the gearbox weight is the closest result among all of the validations performed. The gearbox weight in the Romax Concept (Student Edition) is 357.8 kg, and the Python result for the gearbox—excluding the casing—is 364 kg, a difference of 6.2 kg. The design-rating result also indicates success, as is the safety factor shown in Figure 8.10.

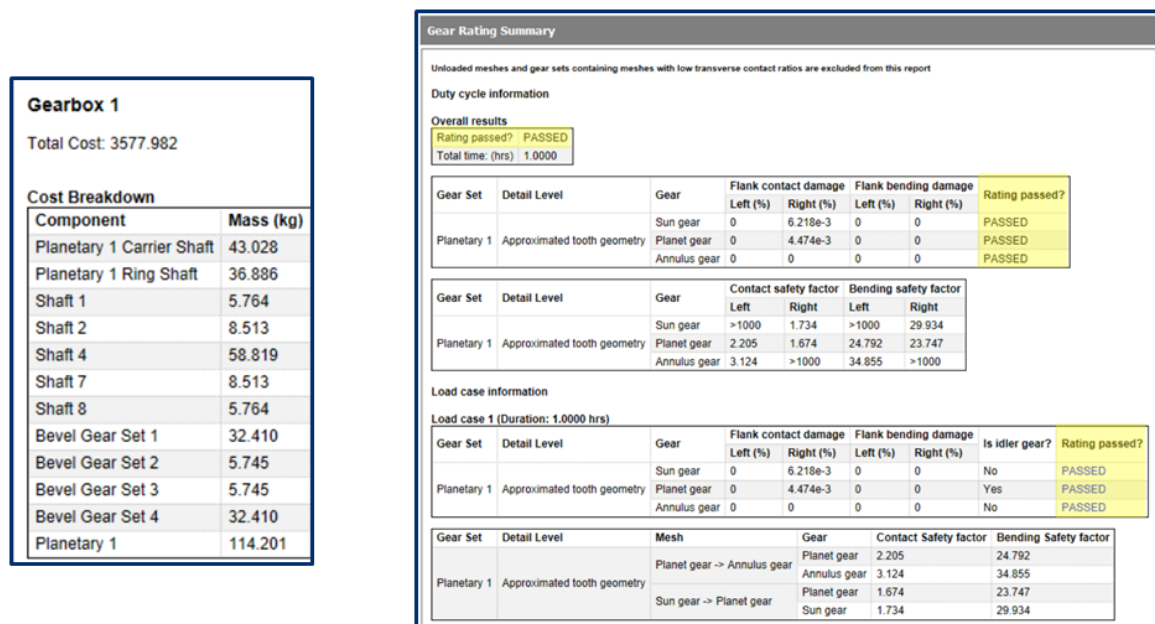


Figure 8.10: Gearbox weight and sizing, and safety factor result

Chapter 9: Results

9.1 Physics-Based Weight Calculation

The foundational step of this project involved converting traditional analytical gearbox weight calculations into a robust Python script. The specific architecture studied in this project is a dual-stage helicopter main rotor gearbox. The powertrain consists of a high-speed input spiral bevel gear stage (which changes the axis of rotation from horizontal to vertical) followed by a low-speed, high-torque epicyclic planetary stage driving the main rotor mast.

To accurately model the complex physics of a helicopter main rotor gearbox, the script accepts 31 independent input parameters. These inputs were chosen based on typical, realistic ranges utilized in the modern aerospace industry. Key inputs include:

- Fundamental gear geometry (helix angle β , normal pressure angle α_n , face-width to diameter ratio λ_d)
- Planetary configuration (number of planets N_p)
- Casing clearances
- Material properties

From these inputs, the script calculates the initial geometry of the entire gearbox assembly. It sequentially evaluates the required physical dimensions for the input bevel gears, the planetary gear stages, the shafts, and the outer housing required to safely and reliably transmit the power without yielding contact fatigue or bending stresses.

A critical design parameter in dual-stage gearboxes is the intermediate shaft speed, which perfectly balances the torque and weight distribution between the high-speed input bevel gears and the low-speed, high-torque planetary stage. To find this optimum intermediate speed, a Sweep Search Algorithm was utilized.

To ensure the physical validity of the initial sizing calculations, the foundational gearbox configuration was verified in Romax where the simplified analytical equations used in Python model accurately represent sound aerospace designs.

9.2 Sensitivity Study

A sensitivity study is used to quantify the absolute importance of design variables and understand their direct, often non-linear, effect on system weight and physical size. The input variables were systematically varied to study their effect within practical, manufacturable ranges. Through this sensitivity study, the primary driving variables were identified that dictate most of the gearbox's mass as shown in Table 9.1 and Table 9.2.

Table 9.1: Sensitivity Analysis Result for the weight

Rank	Variables	Delta
1	α_n	1.8405
2	P	0.8069
3	λ_d	0.6802
4	β	0.5392
5	ρ_{shaft}	0.5344
6	σ_{shaft}	-0.4344
7	n_{rotor}	-0.4324
8	N_p	-0.4109
9	K_a	0.4064
10	K_t	0.4064

Table 9.2: Sensitivity Analysis Result for the size

Rank	Variables	Delta
1	α_n	2.4391
2	N_p	-1.9291
3	β	0.7988
4	P	0.6978
5	λ_d	0.6178
6	σ_{shaft}	-0.3770
7	C_t	0.3015
8	$L_{clearance}$	0.2250
9	$\sigma_{allowable\ contact}$	-0.2219
10	Z_E	0.2210

9.3 Dataset Generation and Material Selection

To train an Artificial Intelligence (AI) model capable of predicting complex physics, we require a massive dataset. The top driving variables identified in the sensitivity study were varied to generate tens of thousands of unique gearbox designs as shown in Figure 9.1

The samples for data generation were generated using Sobol Sequence Sampling rather than traditional Monte Carlo random sampling. Monte Carlo methods tend to cluster data points randomly, leaving massive 'gaps' or 'holes' in the multi-dimensional parameter space. Sobol sequences are quasi-random, low-discrepancy sequences explicitly designed to distribute points as uniformly as mathematically possible across an N-dimensional space. This ensures the ML

model receives a perfectly balanced learning of the edges of the design space just as well as the center. The dataset was generated using 11 distinct material profiles such as: AISI 9310 Steel, Nitrided Steels, Ferrium C61 and Titanium.

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	
beta	P	alpha_n	lambda_d	n_rotor	K_t	K_a	C_c	C_t	K_web	C_w	n_s	N_p	rho_gear	rho_shaft	sigma_all	sigma_all	sigma_sha2_E	n_in	L_clearanc	lambda_b		
14.31029	2414368	21.53206	0.521383	249.9176	1.067976	1.409333	0.260396	4.806229	1.011466	0.552829	2.336531		6	2810	2810	6E+08	2.5E+08	1.5E+08	130000	20000	0.035	0.3
15.99638	1959015	19.81357	0.789481	259.0618	0.945756	1.297713	0.215041	4.430702	1.068228	0.511999	2.034145		6	3200	3200	2.5E+09	4E+08	2E+08	220000	20000	0.035	0.3
18.95423	2191720	20.61893	0.671622	268.6439	0.964876	1.43715	0.238029	4.087657	1.079899	0.547502	2.23494		6	7850	7850	1.65E+09	4.8E+08	1.5E+08	190000	20000	0.035	0.3
12.12737	2307823	19.07991	0.879145	246.7234	1.077382	1.470208	0.231575	4.241675	1.060974	0.528729	2.438873		4	7850	7850	1.65E+09	4.8E+08	1.5E+08	190000	20000	0.035	0.3
10.7392	1712929	18.27553	0.839437	240.2759	1.048907	1.251118	0.286436	4.712268	1.049797	0.587529	2.385717		5	3200	3200	2.5E+09	4E+08	2E+08	220000	20000	0.035	0.3
17.56408	1844658	22.04844	0.609844	277.5864	0.911426	1.208584	0.292495	4.617094	1.018713	0.581259	2.182658		5	4430	4430	1.1E+09	4.5E+08	3E+08	162000	20000	0.035	0.3
12.92213	2075120	21.11574	0.559699	256.2819	1.008275	1.349618	0.209007	4.273137	1.099303	0.518215	2.268575		4	7850	7850	1.65E+09	4.8E+08	1.5E+08	190000	20000	0.035	0.3
17.38654	2546098	17.52237	0.729485	267.9311	0.980481	1.385654	0.266825	4.897642	1.030382	0.571547	2.102937		5	1600	1600	3E+08	4E+08	5E+08	100000	20000	0.035	0.3
13.39268	2144403	18.03882	0.62527	262.2045	1.032593	1.486311	0.229759	4.12595	1.041203	0.524475	2.356143		4	7800	7800	2E+09	6E+08	2.2E+08	195000	20000	0.035	0.3
16.29095	1603173	20.69481	0.857468	250.5626	0.953237	1.225274	0.293828	4.500393	1.085368	0.565286	2.022331		6	3200	3200	2.5E+09	4E+08	2E+08	220000	20000	0.035	0.3
18.66086	2398301	18.95196	0.575523	240.9223	0.935741	1.364863	0.271839	4.96942	1.072135	0.57502	2.192136		6	8190	8190	1.3E+09	5E+08	4E+08	200000	20000	0.035	0.3
10.58477	2529999	20.09791	0.773658	273.2304	1.023222	1.387954	0.252501	4.82625	1.09071	0.59379	2.489551		4	7800	7800	2E+09	6E+08	2.2E+08	195000	20000	0.035	0.3

Figure 9.1: Generated Dataset

9.4 Machine Learning Models

With the data generated, multiple machine learning architectures were investigated to serve as the surrogate model. The goal of a surrogate model is to predict the true physics calculations in microseconds rather than seconds. Different models such as Random Forest, Multi-Layer Perceptron (MLP) and HistGradientBoosting (HGBT) were evaluated. Model performance was measured using the Coefficient of Determination (R^2). R^2 is a statistical measure that represents the proportion of the variance for a dependent variable that's explained by the independent variables. An R^2 of 1.0 indicates flawless, perfect prediction.

Based on benchmarking across various maximum tree depths as shown in Table 9.3, HistGradientBoosting consistently outperformed all other models, achieving a R^2 score of > 0.89 with a Mean Absolute Error (MAE) of less than 42 kg on a 500+ kg gearbox. Therefore, HistGradientBoosting was selected as the core architecture.

Table 9.3: Different Model Performance Comparison

Model Name	Depth	R^2	Mean Error (kg)	Time
Random Forest	10	0.8721	46.66	23.0s
XGBoost	10	0.8886	42.46	45.4s
HistGradientBoosting	10	41.85	10	10.9s
Random Forest	25	0.8823	44.18	9.7s
XGBoost	25	0.8626	46.80	324.5s
HistGradientBoosting	25	41.88	10	4.9s

9.5 The Discontinuity Problem and Model Cascade Architecture

During initial training, a severe physical discontinuity problem was discovered. In real-world manufacturing, gear teeth cannot be cut to any arbitrary decimal size; they must conform to standard tooling sizes known as ISO Modules (e.g., 2.0 mm, 2.5 mm, 3.0 mm). When the continuous physics math predicts a required gear module of 2.1 mm, it must artificially snap up to the 2.5 mm standard. This snapping behavior creates massive, sudden jumps in the final gearbox weight. Standard Machine Learning models fundamentally struggle to learn these sudden discontinuous step-functions, leading to high error rates.

Therefore, instead of training one model to do everything, the brain was splitted into two sequential models:

1. Model 1 (The Sizer): Predicts the continuous, exact mathematical module required by the physics formulas.
2. The Snapper: Python function intercepts Model 1's prediction and perfectly snaps it to the nearest standard ISO module.
3. Model 2 (The Weigher): Takes all the original geometric inputs plus the newly snapped module feature to predict the final Weight and Volume.

This architecture bridges the gap between pure continuous mathematics and discrete manufacturing limitations, eliminating the discontinuity error. This results in improvement in the accuracy of the AI model as shown in Table 9.4.

Table 9.4: Model Cascade Performance

Model Name	R^2	Mean Error
Sizer	0.985	0.0003 mm
Weigher	0.984	15.36 kg

9.6 Feature Importance (Permutation Importance)

Permutation Importance physically breaks the relationship between a specific feature and the target variable by randomly shuffling that column's data. We then measure how severely the model's R^2 accuracy drops. A massive drop in accuracy indicates that the model relies heavily on that feature.

The results confirmed our physics models: the physical size of the gear teeth was the single most critical feature, followed closely by the geometric knobs (β , α_n , λ_d) and the planetary configuration (N_p) as shown in Figure 9.2 and Figure 9.3. These specific features therefore can be selected as the primary variables for the final optimization phase.

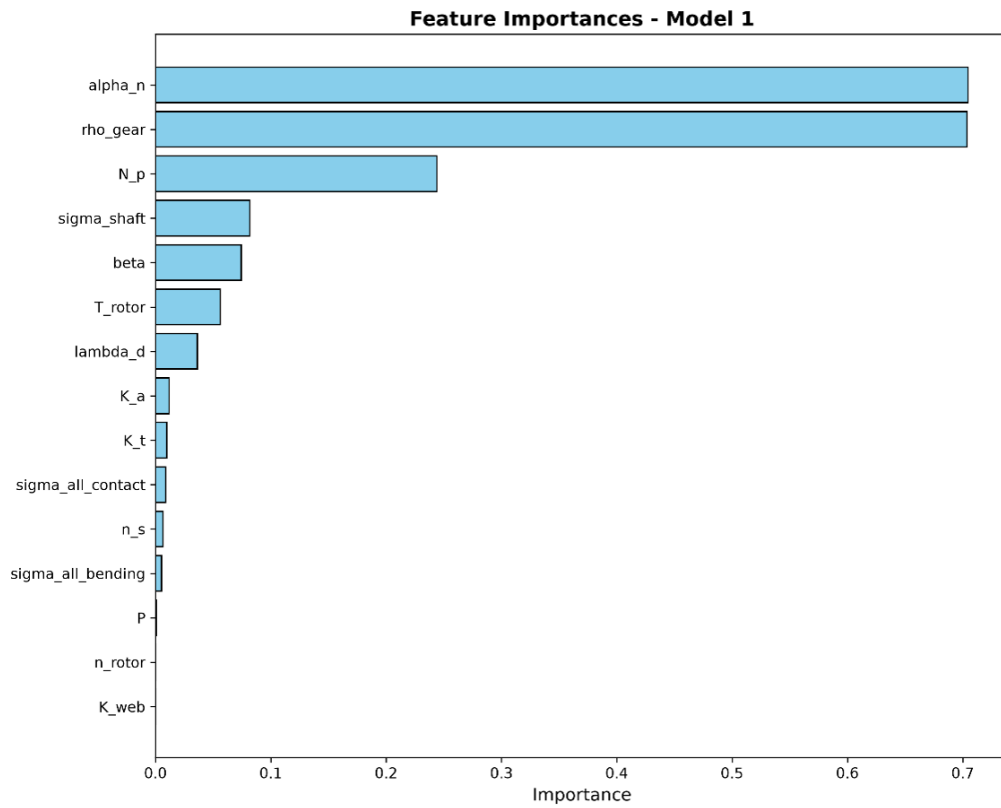


Figure 9.2: Feature importance of the sizer

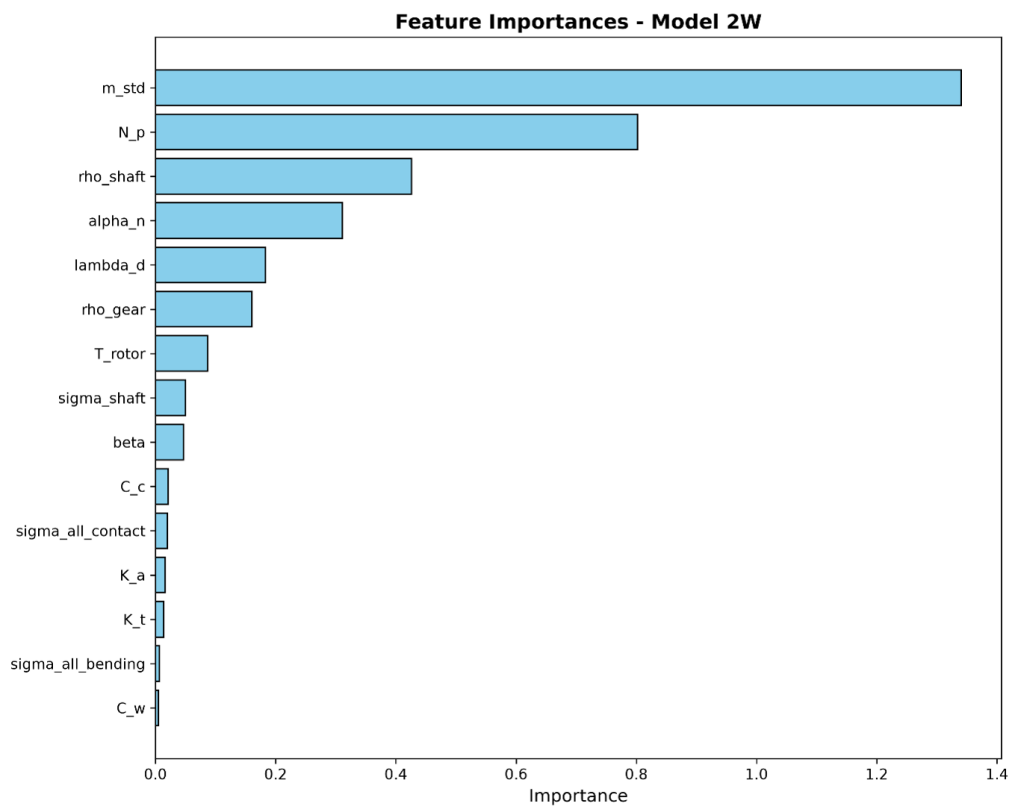


Figure 9.3: Feature importance of the weigher

Chapter 10: Conclusion

10.1 Lessons Learnt

- Studied helicopter gearbox topologies, including those used in the A.R.T. program, and compared stage-layout options, gear-pairing strategies, and certification constraints to see how design decisions shape overall geometry and mass.
- Observed how varying single operational parameters causes complex changes across the entire gearbox design and that mass does not scale linearly with torque or input power.
- Learnt that machine learning models designed for continuous spaces naturally fail to approximate step-functions.
- Utilized model-agnostic feature evaluation to rank which operational parameters truly dominate final mass and size.
- Learned Romax Concept (Student Edition) in depth: how it works, creating gearbox configurations manually, running simulations, and learning about safety factors and design capabilities.
- Gained hands-on experience using Romax Concept (Student Edition) to build, simulate, and validate complex gear arrangements.
- Learnt range of machine-learning models (linear regression, random forests, gradient-boosted trees, and neural networks) evaluating their strengths and limits for predicting size, weight, and performance.
- Learnt how to distribute points in a N-dimensional space and why completely random sampling leaves blind spots for the AI, and how quasi-random Sobol Sequences solve this by spacing out training points uniformly.

10.2 Conclusion

The workflow introduced in this work replaces the traditional, time-consuming, and skill-dependent gearbox design cycle with a repeatable, data-driven process that still respects every engineering constraint. The sizing engine, implemented in Python, accepts 31 design, material, and operating variables and returns the full geometric description of a dual-stage helicopter main gearbox (spiral-bevel input and planetary output).

The design parameters produced by the Python design script were manually entered into the Romax Concept (Student Edition). The software then returned the gearbox weight, the stress-analysis results, the actual gear ratio, and the safety factor. The Python output predicts a gear

mass of 364 kg (excluding the casing), which is within $<2\%$ of the 357.8 kg mass reported by the Romax Concept (Student Edition). A sensitivity study identified gear modules, pressure and helix angles, and planetary configuration as the most influential parameters for weight. These results informed the sampling strategy for the training and highlighted the location of the dominant design trade-offs.

A Sobol-sequence generated tens of thousands of distinct configurations to train two surrogate AI models. A first-stage “Sizer” (HistGradientBoosting) predicts the continuous gear module; the output is then snapped to the nearest ISO standard module. A second-stage “Weigher” (also HistGradientBoosting) predicts total mass and volume from the snapped module and the remaining design variables.

The cascade approach eliminates the discontinuity that appears if a single model is trained on the raw, continuous outputs. The final surrogate achieves ($R^2 = 0.984$) and a mean absolute error of 15.36 kg.

Future work will extend the database to cover a wider range of transmission configurations and validate the predicted designs through prototype testing. The framework presented here provides a scalable foundation for rapid, physics-based design of high-performance gearboxes in aerospace applications.

Chapter 11: Recommendations

11.1 Next Steps

- **Integrate Shock Load limits:** The current physics solver only checks steady-state stress limits. The team recommend expanding the sizing equations to include transient shock loads. This will help the model to reject brittle materials early on.
- **Physics Based ML model:** The traditional ML models only look at numeric correlations. If they extrapolate, they can predict impossible designs. So, the stress limit can be incorporated into the model to prevent impossible solutions.
- **Shaft and Housing Deflection:** The analytical model assumes casing and shafts are rigid. However, shafts can deflect under heavy torque, causing the gear teeth to misalign. So, the team recommend exporting data from Romax to model shaft deflections and handle teeth misalignment.
- **Romax Concept (Student Edition):** This version is the trial version of the software, so the '1 main bevel gear to 2 side bevel gears' is not possible; the team created a similar configuration with two main bevel gears stacked on top of each other to observe gearbox performance. The planetary gear modules in this edition have detailed design limitations, yet all tests have shown the system to operate without abnormal behaviour, making it suitable for preliminary validation studies.

Bibliography

- [1] Gearboxes of helicopter power plants and turboprops - tutorial kharkiv “khai” 2016, 2016. n.d.
- [2] admin. Fundamentals of gearbox design - covalo industries, feb 2025.
- [3] National Aeronautics, Space Administration, and U.S. Army Research Laboratory. Advanced rotorcraft transmission (a.r.t.) joint program report. Nasa technical memorandum, NASA Glenn Research Center, 1995.
- [4] C. Autore, Luca D’Angelo, D. M. Cirelli, and D. Pier. Performance-driven design of multi-stage gear transmissions for e-drive systems, jun 2026.
- [5] MSC Community. Hexagon.com - romax suite, 2026.
- [6] MSC Community. Site.com - romax concept, 2026.
- [7] R. R. Craig and M. C. Bampton. Coupling of substructures for dynamic analyses. *AIAA Journal*, 6(7):1313–1319, 1968.
- [8] Romax | Electromechanical Drivetrain Design. Cadence.com - romax, 2026.
- [9] European Committee for Standardization. Case hardening steels - technical delivery conditions (en standard no. 10084), 2009.
- [10] International Organization for Standardization. Rolling bearings - methods for calculating the modified reference rating life for universally loaded bearings (iso technical specification no. 16281). Technical report, ISO, 2008. ISO TS 16281.
- [11] G. L. Handler. Advanced rotorcraft transmission (a.r.t.) program: Component test and evaluation phases. Nasa technical memorandum no. 105000 / army research laboratory technical report, NASA Glenn Research Center, 1992.
- [12] Hexagon Manufacturing Intelligence. Leveraging ai/ml technologies in powertrain and transmission design frameworks (romax technology whitepaper series), 2023.
- [13] A. Maciejczyk and Z. Zdziennicki. *Design basic of industrial gear boxes: Calculation and design case example*. PhD thesis, Technical University of Łódź - Department of Vehicles and Fundamentals of Machine Design, Łódź, Poland, 2011. Master’s thesis.
- [14] Nexus Home | Software | Romax. Hexagon.com - nexus, 2026.
- [15] Romax Software | Drivetrain Design & Simulation Tool. Creotek systems india llp - romax, mar 2024.
- [16] T. Ulkan and B. Taban. Gearbox design of an unmanned helicopter. *Review of Gearbox Design of an Unmanned Helicopter*, 2021. Review article - no DOI provided.
- [17] Mert Vardar. Design alternative for a helicopter final stage planetary gearbox | gear solutions magazine, oct 2025.